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Financial Crisis Prediction :

The Critical Slowing Down

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Master 1 Econometrics-Statistics 2024-2025

Abstract

In this dissertation, we study whether critical slowing down indicators can be used to detect a critical transition or not, in particular financial crises. To achieve that, we manipulate several indicators such as AR(1), variance and skewness, the rolling window and the bandwidth. We apply these methods to key financial indices—S&P 500, Nasdaq Composite, Hang Seng, TED Spread, and FTSE 100—to identify early warning signals preceding major crises: Black Monday (1987), the Asian Financial Crisis (1997), the Dot-Com Bubble (2000), and the Global Financial Crisis (2008). Our results show that we can detect clear early warning signals for Black Monday.

Contents

1 Introduction

Critical slowing down (CSD) is a concept used across several fields, including physics, biology, and finance. It refers to a phenomenon where a dynamic system becomes slower to respond to disturbances as it approaches a phase transition point, often linked to abrupt changes or critical transitions. The concept was noticed by H. Haken [6] in synergetic while working on laser physics and nonlinear system dynamics. According to him, a laser beam represents a stationary self-organized system state. It emerges through a state transition where a system shifts from disorder to order. His work establishes a link between various self-organizing systems and highlights their similar behavior near state transitions. In 1972, Kenneth Wilson [14], in his work on renormalization and critical phenomena, developed a framework for understanding phase transitions and critical phenomena, analyzing how system interactions evolve across different scales. Hohenberg and Halperin [7] deepened the theory of renormalization by using the phase separation of a symmetric binary fluid as an example. The authors classify and characterize the different dynamic universality classes, grouping systems with similar critical behavior laws, while establishing a systematic theory of dynamic critical exponents.

In the ecological domain, the critical slowing down concept has been widely explored. The works of Dakos, Scheffer [2] [3] [9] [10], and their collaborators have demonstrated that critical slowing down can serve as an early-warning signal for predicting abrupt transitions in ecosystems, such as abrupt climate change. Furthermore, Dakos et al. [8] [4] expanded on methods for detecting early warning signals in ecological time series, showcasing the potential of these approaches for understanding critical ecological transitions. As a result, it has attracted growing interest from professionals in various fields, including finance.

One of the first finance bubble that exploded was the tulip mania. The price of bulbs escalated. Tulips were sold for approximately 10,000 guilders at the height of the bubble, equal to the value of a mansion on the Amsterdam Grand Canal. The rapid acceleration began in 1634, followed by a dramatic collapse in February 1637. However, one of the most impactful crises on the global economy was the Wall Street Crash of 1929, which led to the Great Depression, lasting for over a decade. In more recent times, the 2008 financial crisis caused significant economic damage such as a decline by 2.2 % in global GDP in 2009. Given the consequences of such events, researchers have begun to find interest in critical slowing down since it can predict when a system transitions from a state to another. Didier Sornette [11] was one of the pioneers in applying the concept of critical slowing down to financial systems to analyze behaviors before and after market crashes. Similarly, Jean-Philippe Bouchaud [1], in his work on power-laws in financial markets, emphasized the role of critical slowing down mechanisms in detecting economic crises. Other papers, such as those by Sornette et al. [12] [13], have also explored the precursors of these events in financial contexts, developing mathematical models to understand and predict economic crises.

Thus, based on this theory, we could detect early warning signals in financial indices. This is what we are trying to do in this dissertation, by applying advanced mathematical models and statistical methods to identify patterns and potential precursors of financial crises. Therefore, the focus of our paper is: Can we detect early warning signals with critical slowing down? We mainly used the methodology of Diks, Hommes and Wang [5]. This dissertation is organized in four sections. The section 2 covers the critical slowing down theory. The section 3 is dedicated to our data, followed by a section 4 that explains our methodology and how we chose our parameters. Finally, in section 5 we present our results and analyze them to conclude.

2 Critical transitions

We observe a critical transition when there is a bifurcation. A bifurcation is an abrupt change in the state of a dynamical complex as ecosystems, climate or financial crises. It occurs when the parameters take a critical value that we can call a bifurcation point. In a critical transition, there are three equilibriums : two stable and one unstable [5]. The system starts at an upper stable equilibrium. As the control parameter ρ increases, the dynamics of the system change, and the two equilibrium branches (stable and unstable) approach each other. At a certain critical value of ρ, the stable and unstable equilibria meet and annihilate each other. That means, both equilibriums disappear, forcing the system to transition to a new state. This is known as a saddle node bifurcation.

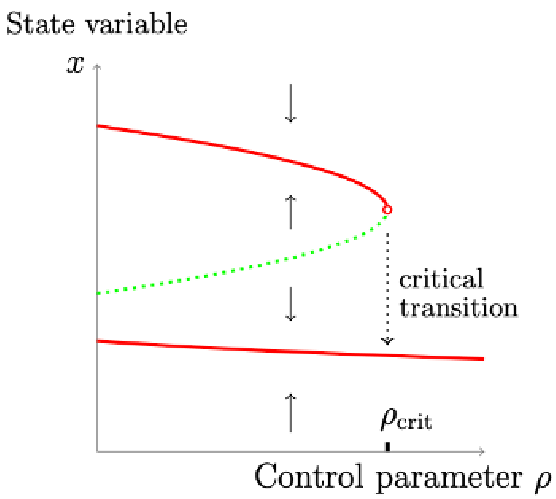

Figure 1: Dynamics of a two-stable-state system and critical transition

In figure 1, there is a saddle-node bifurcation illustrated here. The red lines represent the stable equilibria of the state variable x as a function of the control parameter ρ_{crit}, while the green dotted line indicates an unstable equilibrium. The solid arrows show the direction of movement for the state variable x in the state space, depending on both the control parameter and the current state of x. The stable upper equilibrium branch intersects with the unstable equilibrium branch at the critical parameter value ρ_{crit} . If the system initially resides in the upper stable equilibrium and the control parameter ρcrit is gradually increased, either continuously or in discrete steps, beyond ρ_{crit} , a critical transition occurs, driving the state variable to the lower stable equilibrium [5].

State variable dynamics, like the stock price for a financial market, are affected by a control parameter. The control parameter is either constant or moving slowly. If the stock price is the state variable, we could imagine, as the control parameter, the economic growth. The latter is long phenomena that will affect the stock price, that is considered as a fast variable compared to economic growth [5]. Financial markets are considered complex systems, which is why we can apply the concept of CSD to financial crises [1] [11] [13].

Empirically, critical slowing down can be observed by examining how the system behaves as it approaches the bifurcation point. When the control parameter gets closer to its critical value, the model takes more time to recover until it doesn't recover at all [2]. It is what we called the critical slowing down. Indicators such as autocorrelation and variance of state variables provide clues to this slowing down process [3]. The slower recovery means that the effect of any disturbance persists for more time, making the system's current state strongly influenced by its past states. This persistence leads to higher autocorrelation: the system remembers its previous states more strongly as it approaches the critical transition. Autocorrelation measures how much a system's current state is related to its previous state. A high autocorrelation means that the system's state changes slowly, and its current state depends strongly on its past. On the contrary, a low autocorrelation implies that it changes quickly, and that the system's state is less connected to its past state. The system loses efficiency, and accumulative small perturbations can trigger the system's shift to a critical transition.

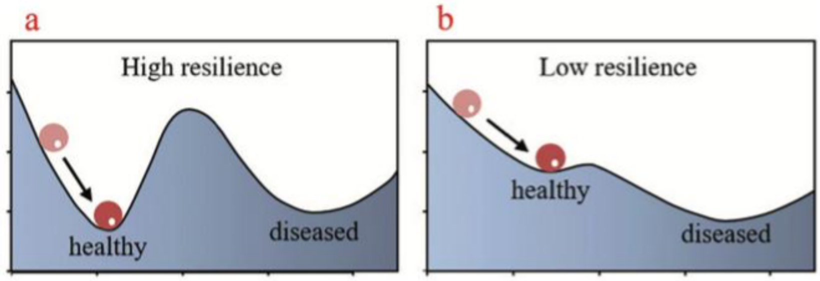

Figure 2: System failover mechanisms

In the figure 2, the ball represents the present state of the system. When there is a high resilience, we are far from bifurcation, there are small variance and fast recovery. In low resilience, we are nearer from a bifurcation, the system takes longer to recover from a shock.

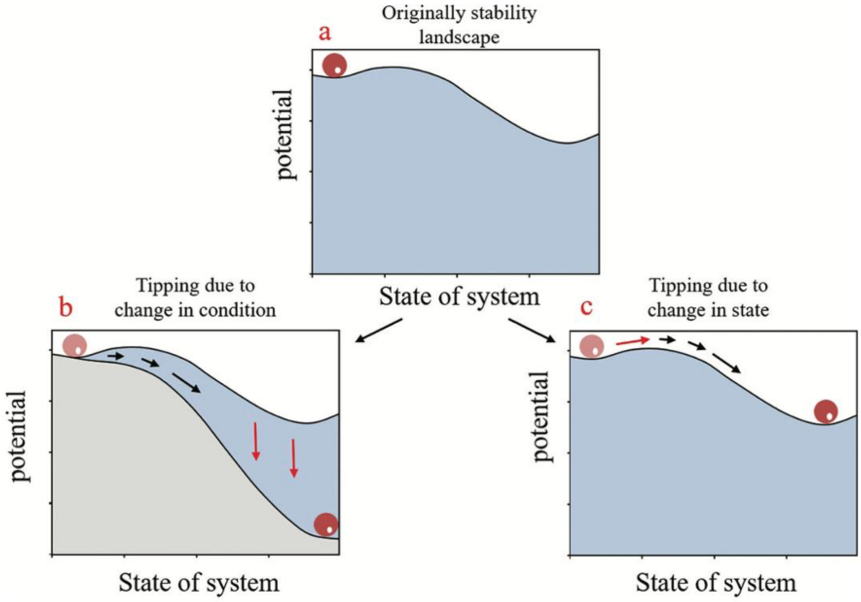

Figure 3: System resilience and recovery

The figure 3 illustrates system resilience and recovery through three scenarios. The initial state (a) represents a stability landscape with a local equilibrium. The system can then tip either due to a change in conditions (b), where the landscape itself shifts, or due to a change in state (c), where a disturbance pushes the system out of its stability basin [10].

The point of this paper is to detect early warning signal of these critical transitions. To do that, we used univariate time series and several indicators to help to prevent the critical transitions: an indicator of autocorrelation such as AR(1), the variance and the skewness [9].

- **AR(1) Process:** An AR(1) process is a time series model where the current value depends on the previous value of the series, with some added random noise. As mentioned previously, we would expect an increase in this indicator as the tipping point approaches.
- **Variance:** Variance measures the average squared deviation of a random variable from its mean. It quantifies the spread or dispersion of a data set or distribution and indicates how much the values fluctuate around their mean over time. We expect an increase in variance near a tipping point, as the system becomes less stable and have more fluctuations.
- **Skewness:** Skewness measures the asymmetry of a distribution. It shows whether the data are skewed to the left (negative skewness) or to the right (positive skewness), or if the distribution is symmetric (zero skewness). As the system nears the tipping point, the stability of the equilibrium changes, potentially causing an asymmetrical distribution. This could result in a higher skewness in absolute value, reflecting the growing imbalance in the system.

In the next chapters, we will see if these indicators can anticipate critical transitions in financial markets, thus financial crises.

3 Data

We used financial crises to test the hypothesis that the indicators mentioned above can anticipate critical transitions. To do this, we studied four major crises: Black Monday [12], the Asian crisis, the dot-com bubble and the 2008 global financial crisis.

To analyse these crises, we selected the most representative and most impacted indices: FTSE 100, S&P 500, TED Spread, Hang Seng and Nasdaq. The necessary data was extracted from Yahoo Finance using the "yfinance" Python library, which enabled us to obtain the daily closing values for each index.

For each of these crises, we determined the date of the critical point. To determine this latter one, we did some research to find a particular date for each crisis and the indices used to represent them.

- **For Black Monday, we took October 13th, 1987.** Black Monday, which occurred on October 19, 1987, is one of the largest stock market crashes in history, with a 22.6 % drop in the Dow Jones Industrial Average in a single day. However, October 13, 1987, represents a time when the markets had already started to show signs of vulnerability. At that moment, global stock indices began to display increased volatility, which is often an indicator of critical slowing down.
- **For the Asian Crisis, we referred to October 3rd, 1997.** The Asian crisis was largely triggered by the devaluation of the Thai baht on July 2, 1997. However, the crisis intensified and began to spread in the following weeks, leading to the involvement of the IMF.
- **For the dot-com bubble, we chose March 15th, 2000.** This date represents the point when the technology stock market reached its peak (the Nasdaq Composite reached a record level of 5,048.62 points on this date), before collapsing. From this date, technology stock indices quickly lost value.
- **Finally, concerning the 2008 financial crisis, we took the Lehman Brothers' bankruptcy.** The decline truly began in early 2008, when several major financial institutions started to report massive losses due to their exposure to subprime mortgages. By March 2008, the S&P 500 had already fallen to around 1,273 points, marking an 18 % drop from its peak. The situation worsened at the end of summer 2008, following the bankruptcy of Lehman Brothers on September 15, 2008, a triggering event for the global crisis. We used the same date for our other indices for this crisis.

Label	Crisis	Critical point	Time series
(1)	Black Monday	13 oct. 1987	S&P 500
(2)	Asian Crisis	3 oct. 1997	Hang Seng
(3)	Dot-com Bubble	15 mars 2000	NASDAQ Composite
(4)	2008 Financial Crisis	15 sept. 2008	S&P 500
(5)	2008 Financial Crisis	15 sept. 2008	TED Spread
(6)	2008 Financial Crisis	15 sept. 2008	FTSE 100

Table 1: Crisis Overview

4 Methodology

4.1 Detrending with Gaussian Kernel

The first step to study the statistics is to detrend our time series. The point of detrending is to remove the trend pattern from the original time series. Otherwise, it can mask or distort the underlying dynamics of the system studied. For example, if stock prices have been rising steadily over many years, we wouldn't be able to see if there is a slowdown or a sudden drop in the short term, which might be an indicator of an impending crisis. Detrending removes these long-term trends to focus on the deviations from that trend, which are more relevant for early crisis detection.

To achieve detrending, we used Gaussian kernel smoothing [5], a common and effective technique. This approach applies a weighting scheme based on a Gaussian kernel function, ensuring that data points closer to the time of interest are given more weight compared to those farther away. The Gaussian kernel function is defined as :

$$G(x, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-x^2}{2\sigma^2}},$$

where x represents the time difference between the observation of interest and other observations in the window, and σ is the bandwidth, which controls the smoothness of the resulting trend. A smaller σ results in a tighter kernel that gives more weight to nearby points, while a larger σ spreads the weights more broadly.

Using this kernel, we calculate the moving average (MA), which represents the smoothed trend of the time series. For a given observation t, the moving average is computed as :

$$\mathrm{MA}_t = \frac{\sum_{r=1}^{T} G(r-t) z_r}{\sum_{r=1}^{T} G(r-t)}, \; t = 1, \ldots, T,$$

where z_r is the original time series value at time r and $G(r-t)$ is the Gaussian kernel weight based on the time difference $r-t$.

After calculating the moving average, we obtain the detrended fluctuations, also called residuals, by subtracting the moving average from the original time series :

$$y_t = z_t - \mathrm{MA}_t, \quad t = 1, \ldots, T.$$

These residuals fluctuate around zero by construction, isolating the short-term deviations from the long-term trend. This is crucial for identifying patterns and anomalies that may signal a critical event, such as a financial crisis.

We detrended the logarithm of each of our indices. Below are some examples of how the data looks after detrending.

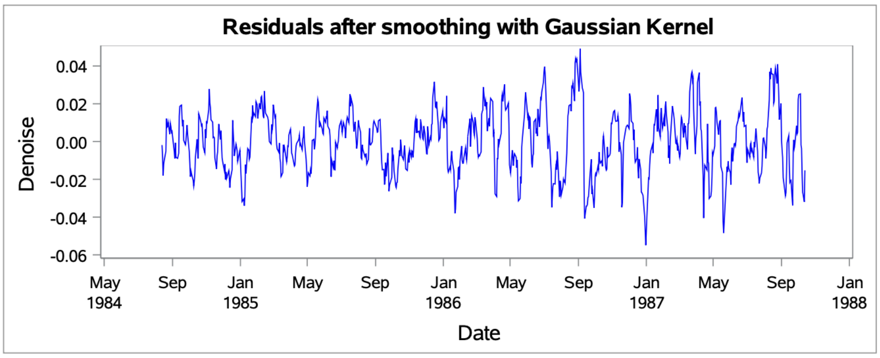

Figure 4: Denoise with $\sigma = 15$ for the Black Monday with SP& 500

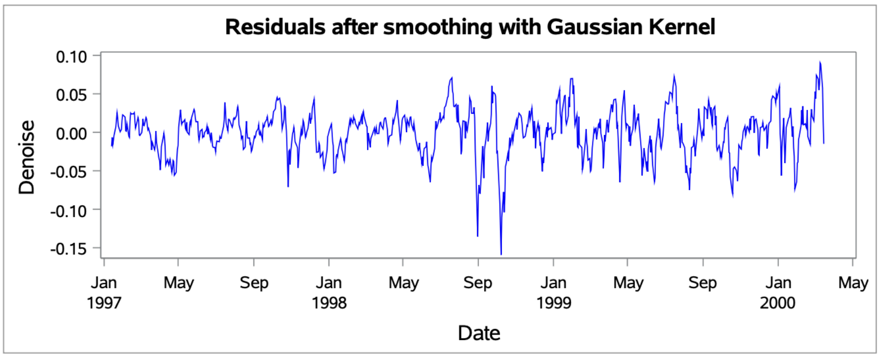

Figure 5: Denoise with $\sigma = 15$ for the Dot-com Bubble with NASDAQ Composite

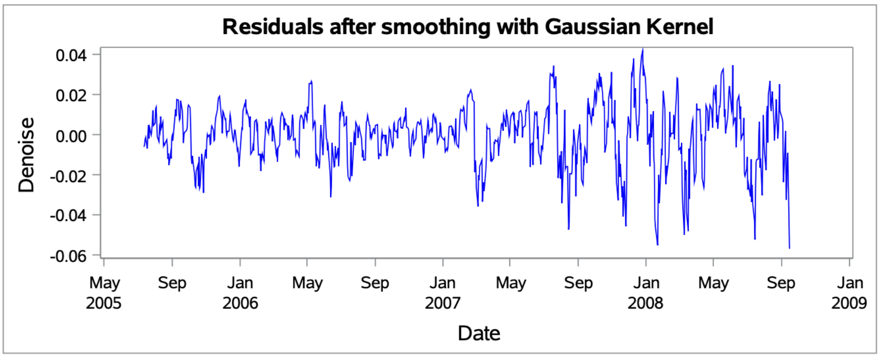

Figure 6: Denoise with $\sigma = 15$ for the 2008 Financial Crisis with S&P 500

To view the other graphs, please refer to the appendix containing the figure 26, figure 27 and figure 28.

4.2 Parameters' choice

For each crisis analyzed, we decided to test several parameters, in particular the size of the sliding window and the sigma smoothing parameter. However, whatever the parameters chosen, the value of T will remain fixed and will always be equal to 800.

As a first step, we have chosen to rely on common practice in the study of CSD. This involves using a sliding window size corresponding to T/2 or T/4, with sigma set at 15 in both cases. A larger window size (T/2) enables us to smooth out the data even more, by integrating older information. However, this reduces the indicator's responsiveness, particularly in the event of imbalances, as recent variations are attenuated by the weight of past data. Conversely, a smaller T/4 window size makes the calculations more responsive, better capturing recent variations, but it can also make the results more volatile. So, we decided to explore both options.

Next, we carried out robustness tests for each crisis to examine all possible parameter combinations and determine which gave the best results for the AR(1) indicator. We use the Kendall to calculate the AR(1) because it's robust to non-normal data distributions and it works well with noisy or nonlinear time series. These tests involved varying sigma between 5 and 20, and the size of the sliding window between 50 and 200. To calculate AR(1), two approaches were considered:

- Compare the initial and final values of the AR(1) by subtracting the former from the latter. A positive result indicates a favorable evolution of the indicator ;
- Focus solely on the final AR(1) value. A high value is essential to conclude that the indicator is capable of predicting a crisis.

We will compare these two approaches to guide our choice of parameters. In addition, we will analyze the evolution of variance by subtracting the initial value from the final value. An increase in variance is additional information, as it validates the indicator only if it evolves at the same time as the AR(1). Both measures must progress together to confirm their usefulness.

Finally, to facilitate interpretation of the robustness test results, we have produced contour plot graphs for simplified, intuitive visualization.

5 Results

5.1 Black Monday

On October 19, 1987, known as Black Monday, stock markets around the world crashed. Dow Jones Industrial Average (DJIA) falling by 22.6 % in one day, the biggest single-day drop in history. This came after months of rising stock prices, creating a bubble, and concerns about a growing trade deficit and a weaker US dollar. By mid-October, markets were already showing signs of trouble, and on October 16, heavy losses occurred during an event called "triple witching," when options and futures contracts expired. Over the weekend, panic spread among investors, and by Monday, markets in Asia, Europe, and the US. all plunged. The crisis showed how connected global financial systems had become, as fear quickly spread worldwide.

5.1.1 T/2

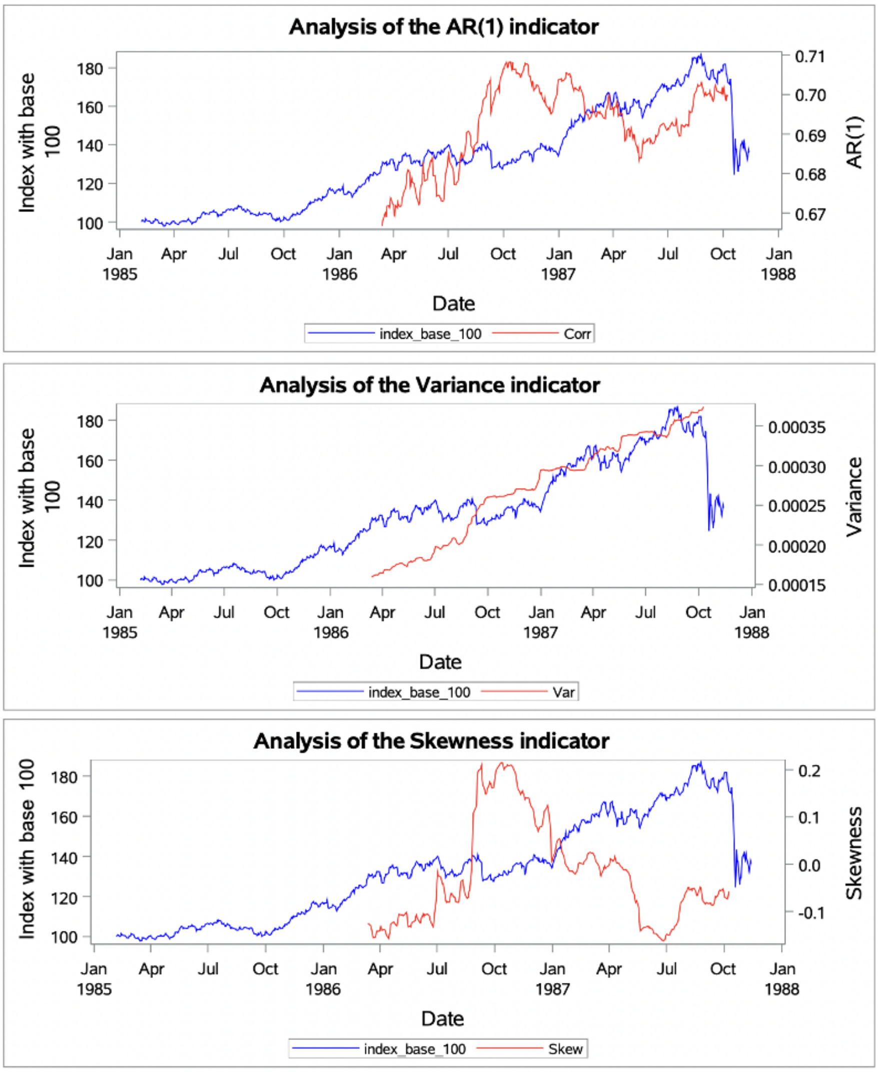

Figure 7: Analysis of indicators for Black Monday with a T/2 window

The AR(1) indicator showed increased volatility and a general upward trend from September 1986, reaching its highest values, close to 0.7, towards the end of the year, between October and December. This gradual rise in the AR(1) could be interpreted as an early warning sign, suggesting that the market was becoming less effective at absorbing shocks. It is notable that these changes occurred before the real stock market crash, suggesting that a critical slowdown was imminent. However, a fall in this indicator was observed in May 1987, perhaps a temporary sign of stabilization before a new plunge into unstable dynamics, illustrating a ‘calm before the storm’. Nevertheless, this fluctuation was part of an overall upward trend, indicating a gradual but critical slowdown.

There has also been a steady increase in variance. This rise in variance is linked to growing disturbances within the system, signaling instability ahead of a critical event, especially as this rise precedes the stock market crash. Combined with the AR(1), the variance shows that the system is becoming increasingly unstable and vulnerable to shocks, foreshadowing a critical downturn.

Asymmetric imbalance before the crash, as measured by Skewness, reflects moderate instability, with fluctuations ranging from 0.2 to -0.1. Skewness rose sharply from August 1987, reaching a peak in October 1987. Moreover, from April 1987, a slight negative trend emerged, probably linked to an unbalanced reaction by investors, often motivated by increased anticipation of potential losses. This imbalance could indicate an imminent crash, although the effects of this instability took some time to become fully apparent in market prices. In short, the system is gradually moving away from its pre-crisis equilibrium point and struggling to return to it.

5.1.2 T/4

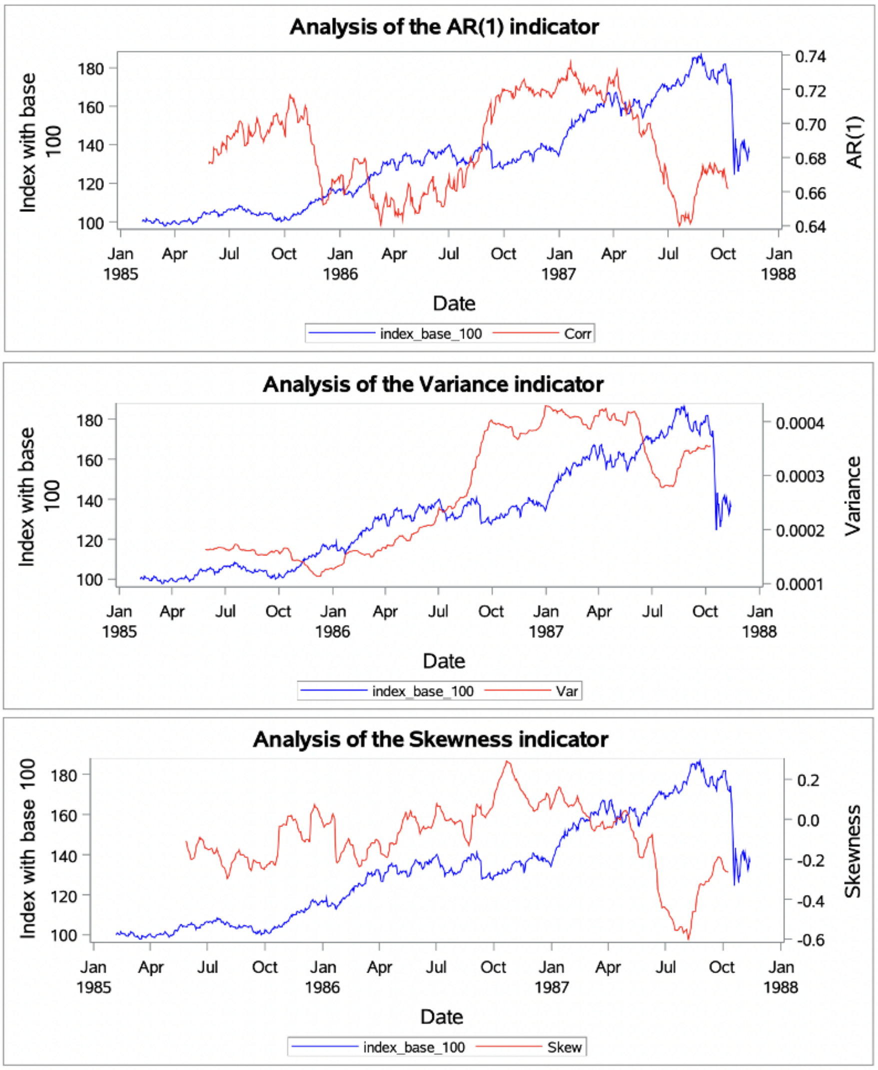

Figure 8: Analysis of indicators for Black Monday with a T/4 window

The AR(1) fluctuates sharply. Initially, it fell around October 1985, but this event was less significant than the sudden rise which began in July 1986. The indicator then reaches around 0.74 in January 1987 and remains at a high level for a prolonged period, before falling sharply after April 1987. This high value of the AR(1) can be interpreted as a signal of a potential crisis. However, this interpretation needs to be qualified, as the indicator had already fallen before and had reached a similar level previously. The sudden fall after April could be explained by an over-reaction of the indicator just before the crisis, followed by a collapse, reflecting behaviour linked to a hysteresis phenomenon, where the market should have fallen at that point, but continues to grow with a time lag.

The variance began to increase in January 1986 and peaked several times between November 1986 and July 1987. However, the combined interpretation of the AR(1) and the variance remains complex because of the ambiguities associated with the behaviour of the AR(1).

Finally, skewness has been moving away from its equilibrium point and into negative territory since June 1987, reaching its lowest value in August 1987, just before the crisis.

5.1.3 Robustness

a) Tests

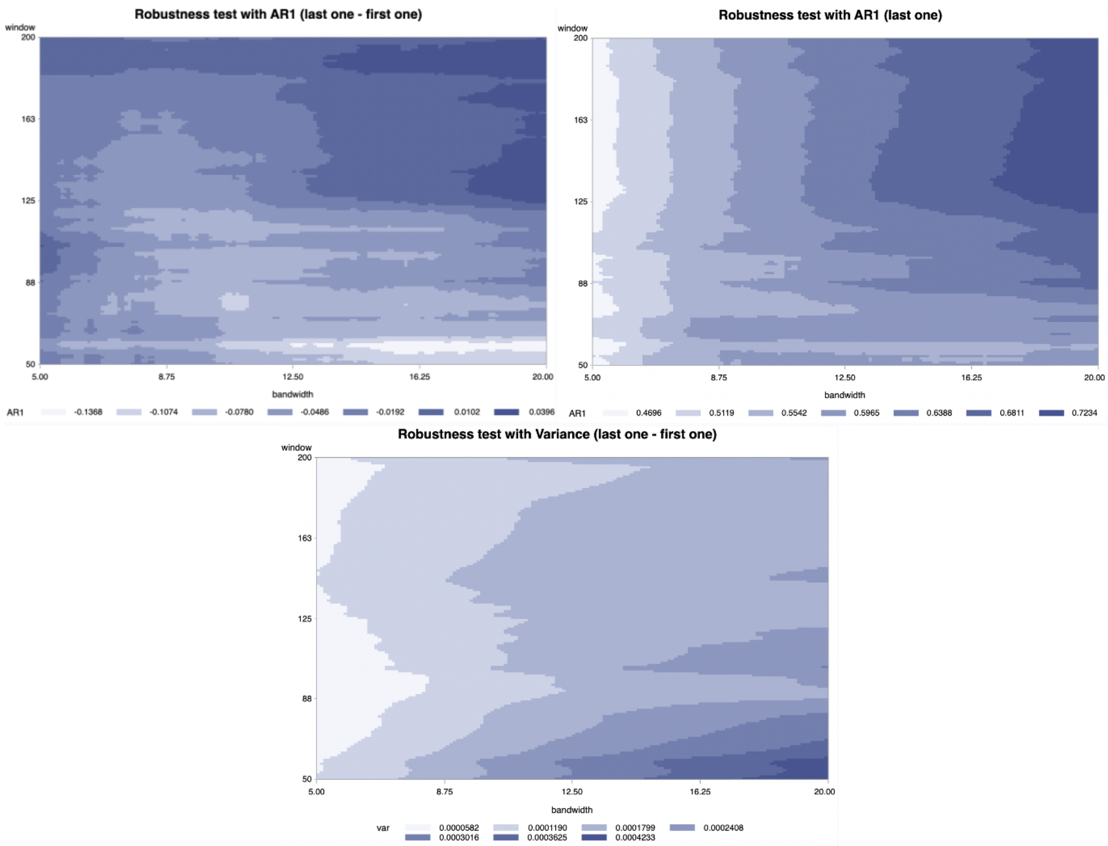

Figure 9: Countour plots for the Black Monday

When analysing these graphs, it is important to take into account the parameters that result in a larger gap between the last and first AR(1), as well as a higher value for the last AR(1), associated with a larger variance. It is by observing these elements that we believe it is more relevant to select $\sigma = 18$ and a sliding window size of 150. We will therefore proceed to analyse the results of our indicators using these new parameters.

b) Interpretations

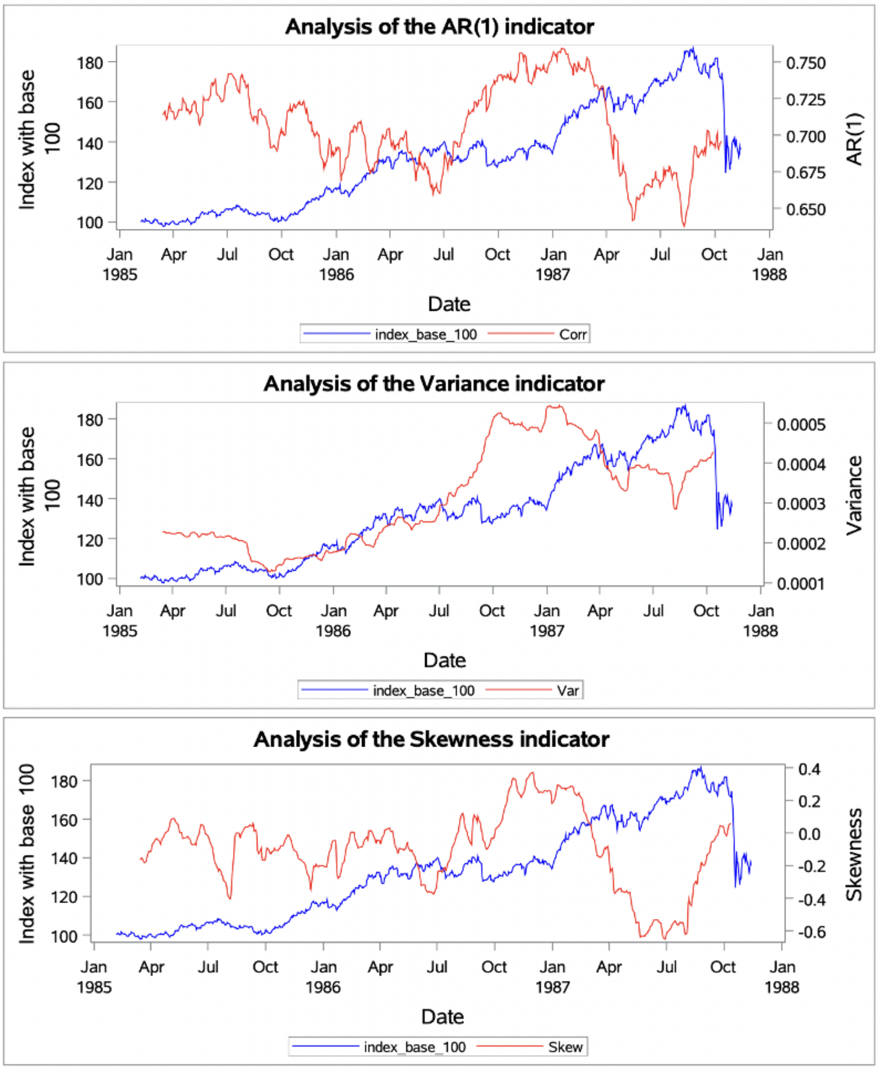

Figure 10: Analysis of indicators for Black Monday with robust parameters

The AR(1) was quite volatile, rising sharply from July 1986 and peaking at around 0.75 between October 1986 and April 1987. It then fell sharply in April 1987, reaching 0.64 in May 1987, just a few months before the stock market crash. As mentioned earlier, this behaviour could be explained by a time lag in the market's reaction.

The variance, meanwhile, shows an overall increase before the stock market crash. The combination of the results for AR(1) and variance can be interpreted as a critical slowdown signal, although this interpretation needs to be qualified because of the ambiguities associated with AR(1).

Finally, skewness shows strong fluctuations, particularly after December 1986, when it struggles to return to its initial point. These fluctuations become more prolonged and pronounced, reaching as low as -0.6 between June and August 1987.

5.1.4 Comparison

In conclusion, the results obtained show that T/4 is particularly useful for capturing imminent changes and detecting critical transitions. However, this configuration remains sensitive to noise and presents a risk of over-reaction, which can lead to false alarms. T/2, on the other hand, offers greater stability and a global view of trends, although it may miss some nearby critical signals.

Robust parameters, on the other hand, bring greater reliability to the results. By using a flexible window size that differs according to the situation, they enable more accurate and consistent analyses to be obtained.

Finally, by comparing the results according to these different parameters, we can see that T/2 is less reactive, which confirms our initial observations. The results between T/4 and the robust parameters remain broadly similar, but the latter provide additional precision. In the context of this crisis, our results can be interpreted as early warning signals.

5.2 Asian Crisis

The crisis began in Thailand in 1997 and quickly spread to East and South Asia. It created a ripple effect, meaning that the impact of the crisis became bigger as it spread from one country to another. The recovery during 1998-1999 was relatively fast. The crisis started when the Thai government was forced to float the baht because it no longer had enough foreign reserves to keep its currency linked to the US dollar.

This led to a sharp drop in the value of the baht, which caused massive financial problems, including capital outflows, falling stock markets, and banking issues. Countries like Indonesia, South Korea, and Malaysia were hit the hardest, with significant drops in their GDP and employment rates. The International Monetary Fund (IMF) provided bailout packages and asked these countries to make economic reforms to recover. While recovery was quick in many areas, the crisis caused serious social and economic damage, such as increased poverty and unemployment. It also highlighted weaknesses in financial systems and led to reforms to prevent future crises

5.2.1 T/2

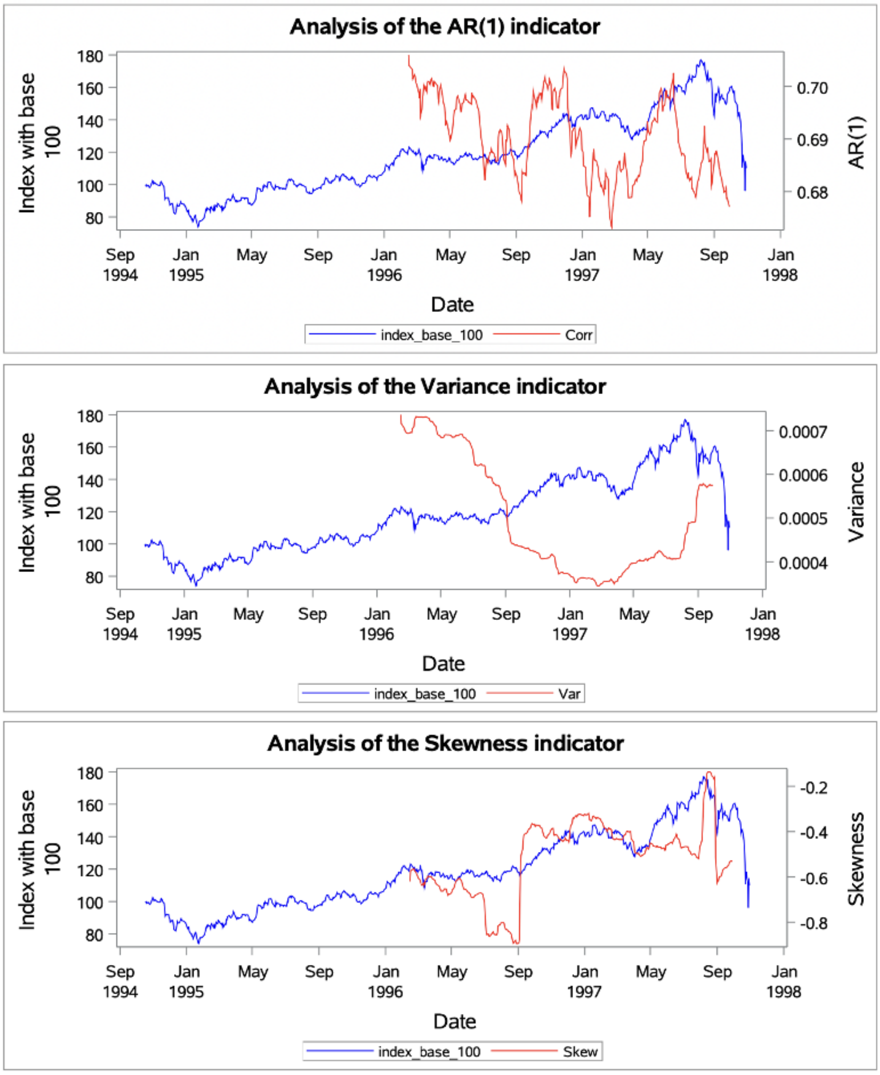

Figure 11: Analysis of indicators for Asian Crisis with a T/2 window

For the AR(1) model, there is no clear upward trend; instead, fluctuations are observed, with rises followed by drops. Regarding variance, it initially declines sharply at the beginning of the correlation curve before rising again in May 1997. As for skewness, there is a noticeable decrease in absolute value, from -0.9 to -0.4 in September 1996. This indicates that the distribution becomes more symmetric and, consequently, more stable. Therefore, these indicators do not provide clear evidence of early warning signals.

5.2.2 T/4

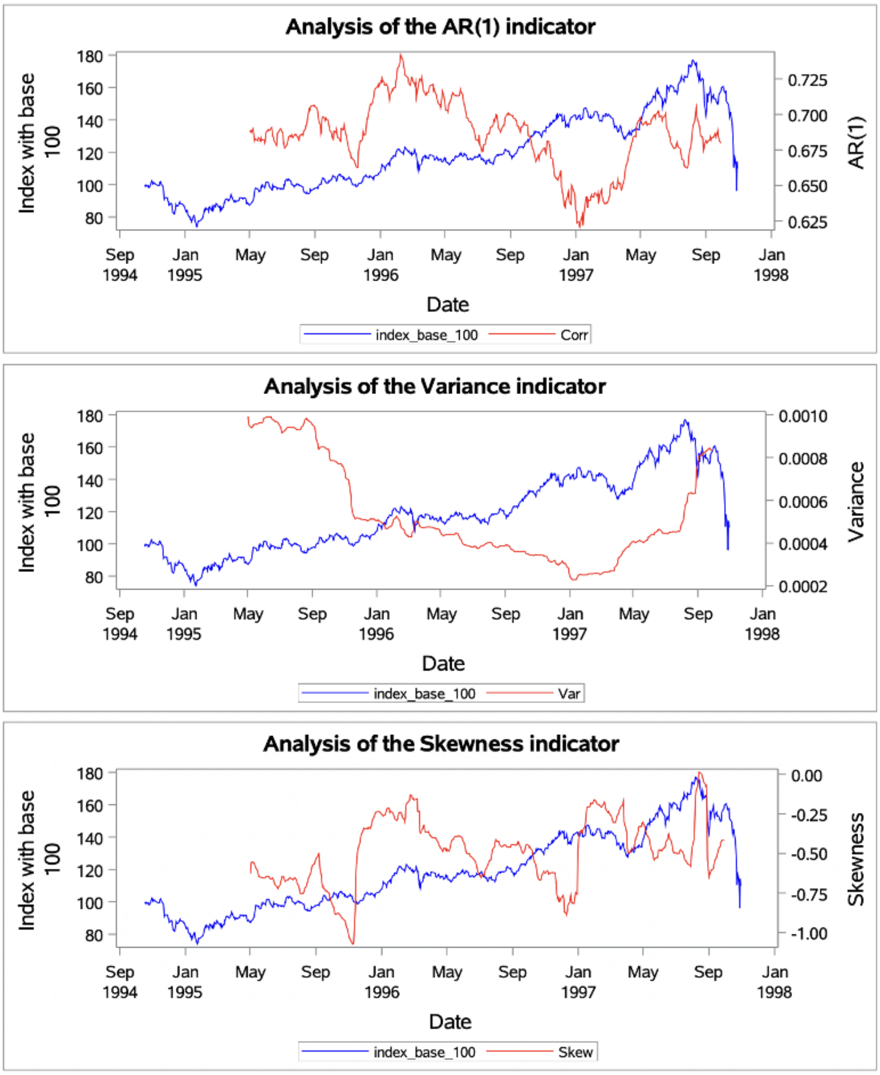

Figure 12: Analysis of indicators for Asian Crisis with a T/4 window

Here, we observe a rise in the AR(1) coefficient from December 1995 (from 0.665 to 0.73), followed by a decline until January 1997 (to 0.625), well before the fall of the index. After that, it rises again to its initial level (between 0.675 and 0.7). This delay in the indicator's reaction could reflect a hysteresis effect, where the crisis takes time to unfold after being triggered. Alternatively, the first rise might be a false positive or false alarm [4], while the second one could represent an early warning signal. Regarding the variance, it only starts increasing in January 1997, rising from 0.0002 to 0.0008, which may indicate heightened instability closer to the event. For the skewness, we observe two pronounced negative peaks: the first in December 1995 (-1) and the second in January 1997 (-0.8). After this date, the indicator gradually trends toward 0, suggesting a shift in the distribution of returns.

5.2.3 Robustness

a) Tests

Based on the figure 29 , we find it more appropriate to select $\sigma = 16.5$ and a sliding window size of 100. We will now proceed to analyze the results of our indicators using these parameters.

b) Interpretations

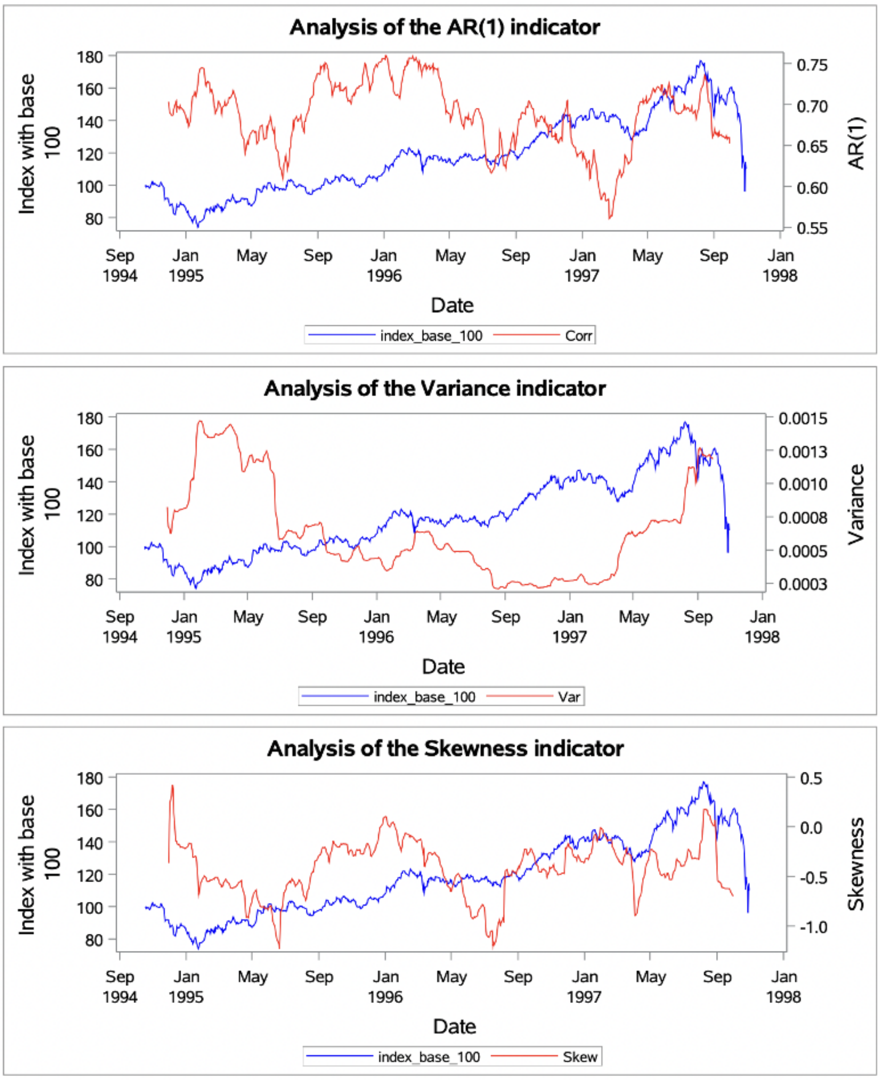

Figure 13: Analysis of indicators for Asian Crisis with robust parameters

For AR(1), we observe the same fluctuations but with more extreme values. The lowest point is 0.55, and the peak is 0.75. For the variance, we can attest to the same increase starting in January 1997, but here, the variance reaches 0.0013. Regarding skewness, there are several fluctuations, but we can identify two notable falls. The first one occurs around July-August 1996, reaching -1.4, and the second one takes place around April 1997, with a value of -0.75. Between these two drops and after the second one, the skewness stabilizes around 0.

5.2.4 Comparison

With window = T/2, there are too few values to detect early warning signals. T/4 and parameters from the robustness test provide more detailed insights, showing fluctuations in AR(1), variance, and skewness. However, neither clearly indicates an impending crisis. T/2 suggests a stabilization of skewness and no strong warning signals. T/4 highlights a temporary rise in AR(1), possibly due to a hysteresis effect, but it remains unclear whether this signals a crisis. The robustness test shows more extreme fluctuations but still no definitive warning signs. In short, a longer window gives more information, but none of the approaches provide clear early warning signals.

5.3 Dot-com Bubble

The dot-com bubble was a stock market bubble that occurred during the late 1990s and reached its peak on March 10th, 2000. This crisis coincided with the rise of the Internet and the World Wide Web. There was significant excitement surrounding start-ups. Investors were enthusiastic about the future of the Internet and poured large amounts of money into new companies. During this period, investments in the NASDAQ composite stock market index rose by 800 %, only to fall by 78 % from its peak by October 2002, erasing all its gains made during the bubble. Most of these companies burned through cash faster than they earned revenue and relied on speculation rather than sustainable business models. As a result, investors sold their stocks, which led to a decline in stock prices. This is why, we use the NASDAQ index to seek for early warning signals.

5.3.1 T/2

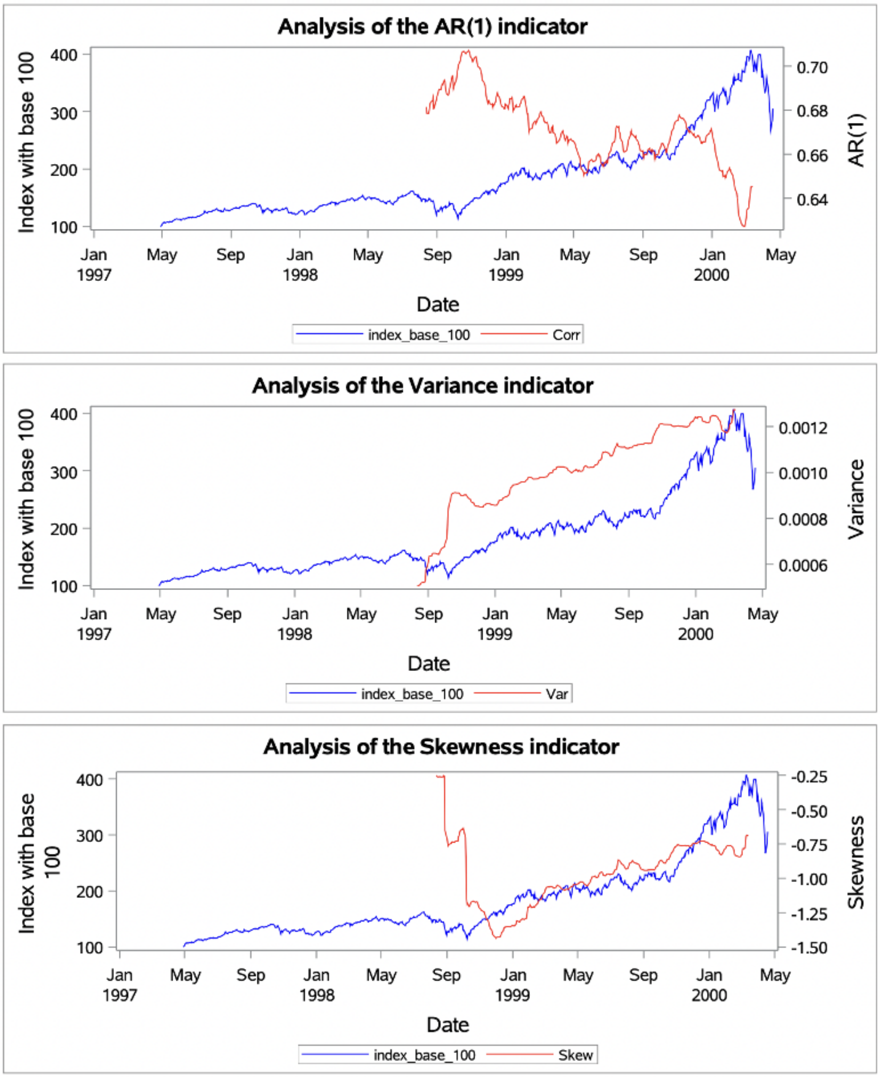

Figure 14: Analysis of indicators for Dot-com Bubble with a T/2 window

For AR(1), we only observe the end of a rise, but we cannot determine where it starts in relation to its initial level. However, we can see that there is an increase in this indicator, showing that the indicator is linked to its previous values. Afterward, there is a fall from 0.71 to 0.62, followed by the beginning of a rise again, possibly returning to its initial level. The increase suggests that the system becomes more dependent on its past values, which could indicate the system is nearing a critical state.

For the variance, we see a constant rise from 0, reaching a peak at 0.0013. This consistent increase suggests growing instability in the market, with higher uncertainty as the period progresses. The increase in variance signals that the market is becoming more volatile, a potential early warning of an impending crisis.

For the skewness, there is a rise in negative values, from -0.25 in August 1998 to about -1.5 in December 1998. This shift reflects an asymmetry in the distribution, with a strong negative skew suggesting a market dominated by lower values, possibly due to panic selling. Afterward, the skewness starts returning toward 0, implying a gradual return to equilibrium.

5.3.2 T/4

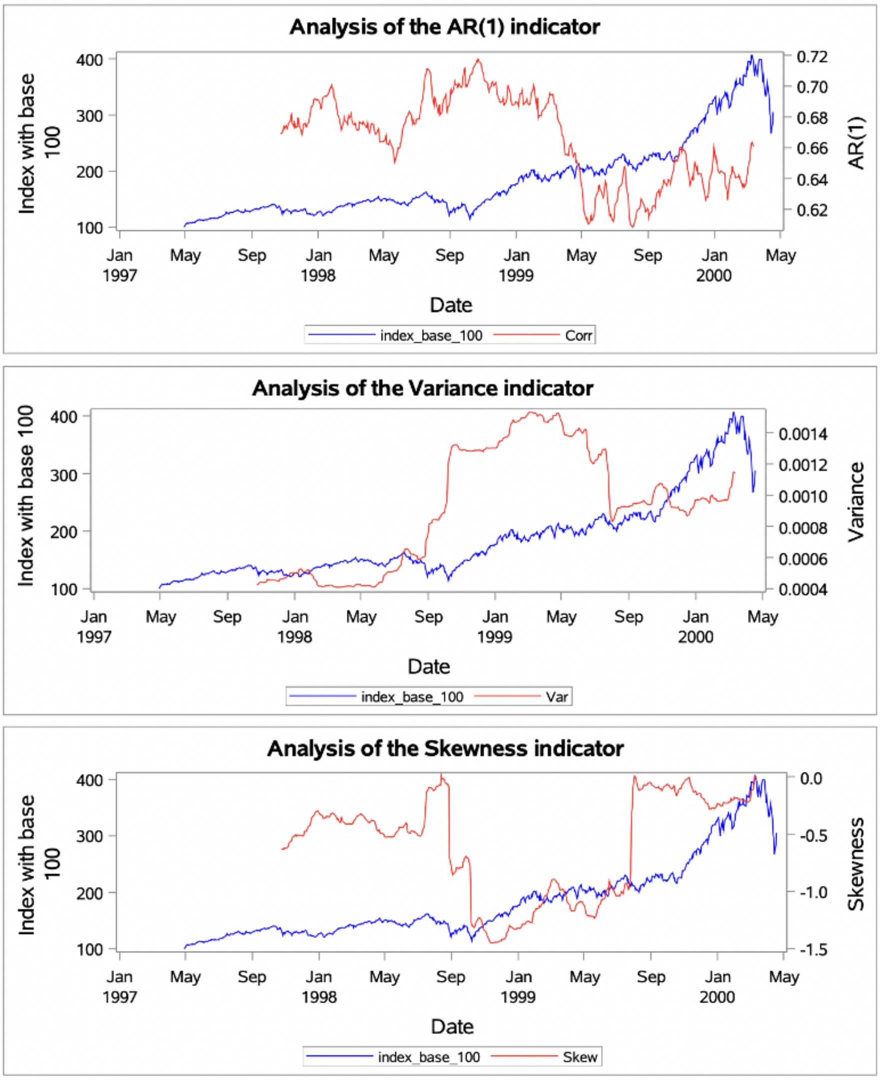

Figure 15: Analysis of indicators for Dot-com Bubble with a T/4 window

For AR(1), we can see an increase from 0.67 on October 1997 to 0.72 on October 1998, followed by a big drop (the value of the indicator was 0.61 on May 1999), and then a small rise that almost made the AR(1) return to its initial value. We could interpret that (the fall) as a false positive. It also could be a sign of hysteresis. That means the bubble has exploded but the effects of the crisis continue to persist, preventing the market from returning to its original equilibrium.

For the variance, there is a large spike from 0.0004 on October 1997 to 0.0016 on April 1999 and then a decrease, but it does not go back to its starting point. After the fall, the variance stabilizes at a level higher than before the initial surge, suggesting that the market has not fully returned to normal conditions.

For the skewness, it increases in absolute value (from 0 to -1.4) and then reaches 0 again, showing a balance in the market distribution. The rise in negative values shows an asymmetry of the distribution to the left so to the low values, that translates a loss of stability.

Combined the three and we can interpret all those fluctuations as an early warning signal. The increase in AR(1) and variance, along with the rise in negative skewness, collectively point to the system's growing instability and its proximity to a critical transition.

5.3.3 Robustness

a) Tests

Based on the figure 30, we find it more appropriate to select $\sigma = 15$ and a sliding window size of 125. We will now proceed to analyze the results of our indicators using these parameters.

b) Interpretations

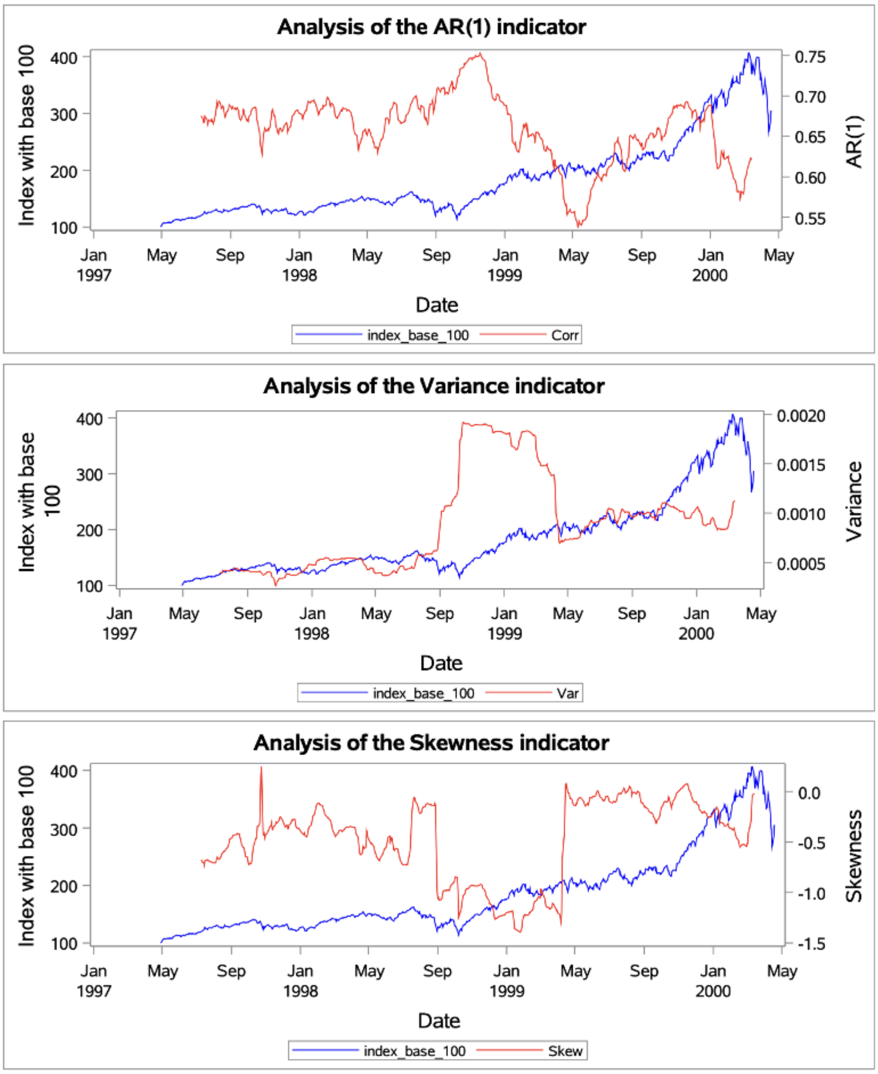

Figure 16: Analysis of indicators for Dot-com Bubble with robust parameters

For the AR(1), the rise begins earlier, reaches a higher level (0.75), and then drops to a lower point (0.55) before fluctuating back to a value similar to the initial one. For the variance, it follows a similar pattern to T/4 but peaks at a higher value (0.0019). Regarding the skewness, the conclusions are consistent with those observed using T/4. Finally, the fluctuations observed when using the robustness test and typical windows T/2 and T/4 are approximately the same. However, the fluctuations are more pronounced. This suggests that personalizing the parameters for each crisis is a more effective approach for detecting early warning signals

5.3.4 Comparison

With T/2, the limited data makes it difficult to detect early warning signals. While AR(1) shows an increase, its starting point is unclear. Variance rises steadily, indicating growing in-

stability, and skewness becomes more negative before stabilizing. However, the trends are less pronounced, making interpretation uncertain.
With T/4, the patterns are clearer. AR(1) rises, drops significantly, then partially recovers, possibly indicating hysteresis. Variance spikes and stabilizes at a higher level, suggesting lasting instability. Skewness also shows a stronger asymmetry before returning to zero. These sharper fluctuations provide stronger evidence of early warning signals compared to T/2.
The robustness test produces similar patterns but with more extreme values. AR(1) fluctuates more dramatically, variance reaches a higher peak (0.0019), and skewness follows the same trend as in T/4. These intensified fluctuations suggest that personalizing parameters for each crisis enhances the detection of early warning signals, making it a more effective approach than fixed window sizes.

5.4 2008 Financial Crisis

The 2007-2008 financial crisis was one of the most severe global financial crises, alongside the 1929 crisis. This crisis was triggered by the collapse of the U.S. housing market. Banks issued risky subprime mortgages to borrowers with poor credit, assuming that housing prices would continue to rise. These loans were bundled into complex financial products, such as mortgage-backed securities (MBS), and sold globally. When housing prices began to fall, borrowers defaulted, causing massive losses for banks and investors. Major financial institutions, such as Lehman Brothers, collapsed, leading to a global credit freeze. Governments intervened with bailouts and stimulus packages to stabilize the economy, but the crisis resulted in widespread unemployment, foreclosures, and a deep recession worldwide. The crisis began in early 2007 and reached its peak with the bankruptcy of Lehman Brothers on September 15th, 2008.

5.4.1 S&P500

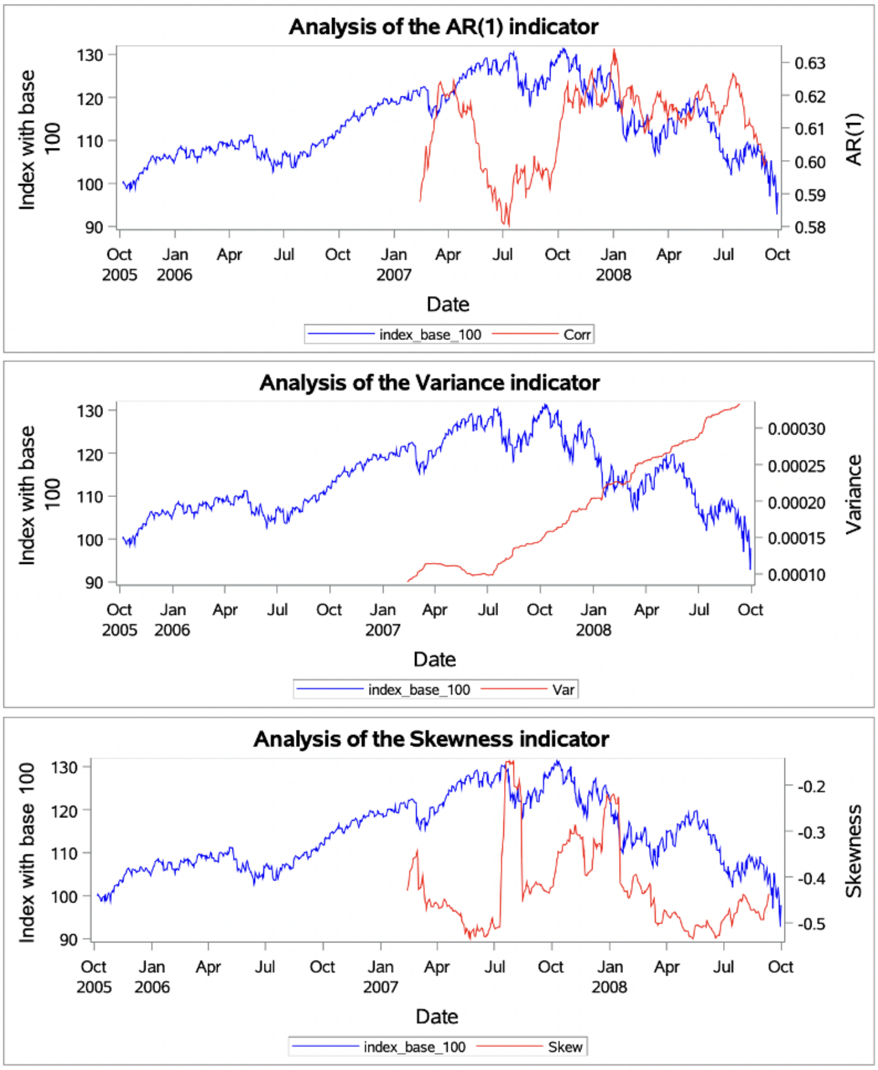

Figure 17: Analysis of indicators for 2008 Financial Crisis with a T/2 window (S&P500)

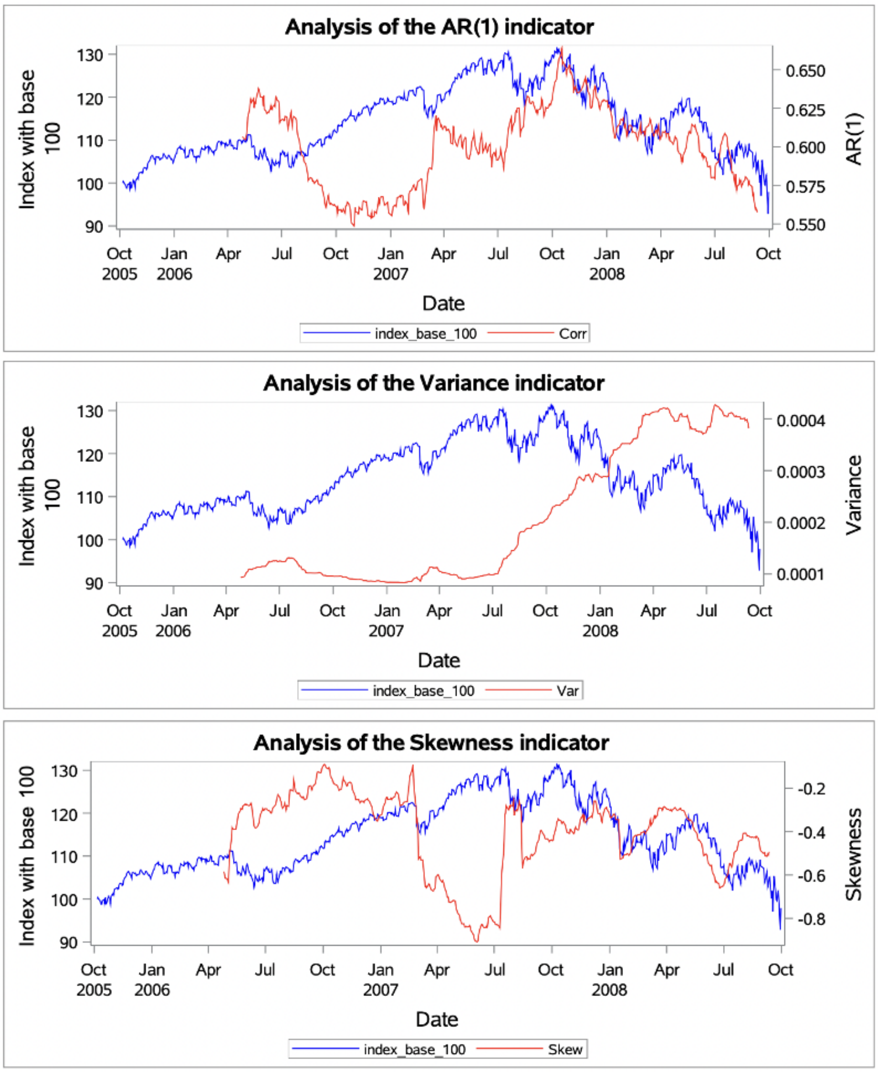

Figure 18: Analysis of indicators for 2008 Financial Crisis with a T/4 window (S&P500)

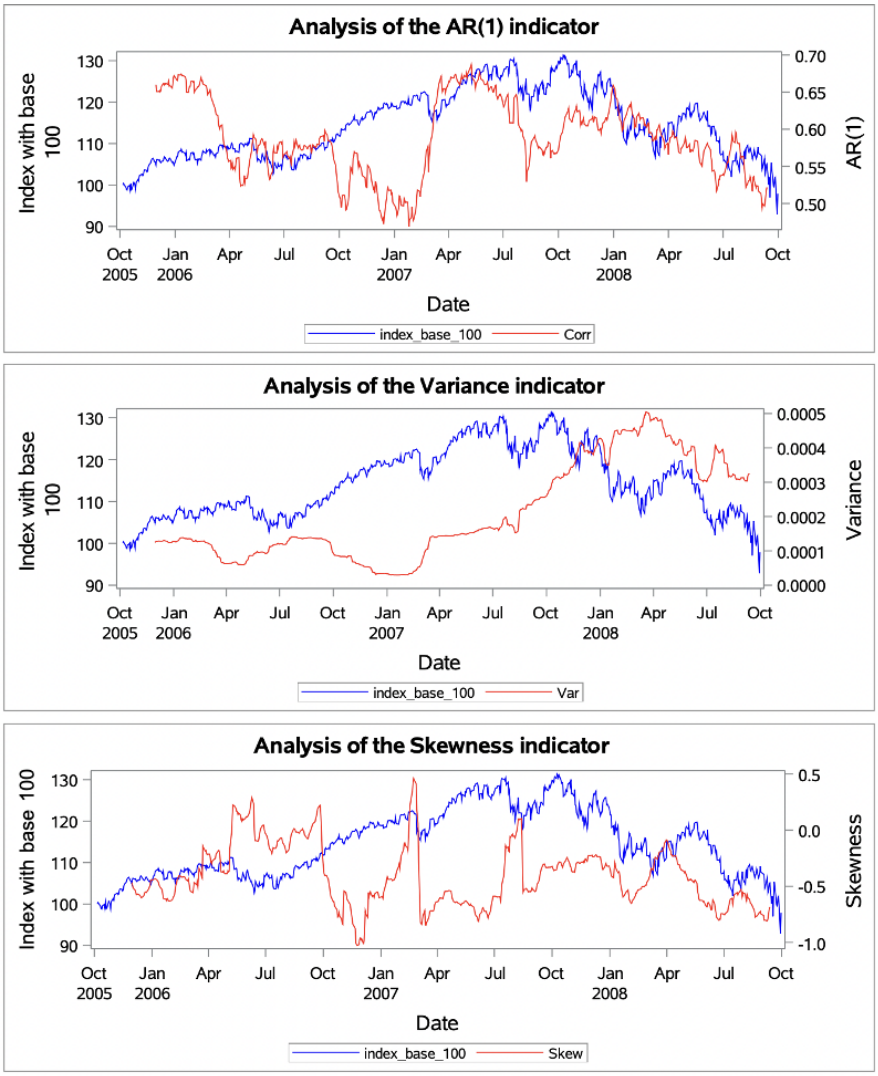

Figure 19: Analysis of indicators for 2008 Financial Crisis with robust parameters (S&P500)

The robustness test (figure 31) led us to select the parameters sigma = 15 and a sliding window size of 100.

On one hand the AR(1) remains highly volatile, oscillating between a minimum of 0.5 and a maximum of 0.7, regardless of the window used. This volatility is particularly noticeable with the robust parameters, which use a smaller window and therefore offer finer precision. Thus, the AR(1) from the robust parameters ought to be a better CSD indicator because it shows an early warning signal with a rise from 0.48 on February 2007 to 0.69 on May 2007.

This critical analysis highlights the major limitations of the AR(1) as a critical slowdown indicator (CSD) in financial markets. Its interpretation requires great caution, as the variations observed do not always correspond to the expected theoretical behaviour. In theory, the AR(1) should increase as a bifurcation point is approached, but the downward phases observed contradict this expectation. This complicates its interpretation: how can we distinguish a real increase, heralding a critical transition, from a simple random fluctuation?

On the other hand, the variance seems to have increased significantly before the stock market crash, which could indicate a relevant signal. However, given the complexity of interpreting the AR(1), it remains difficult to state with certainty that these indicators predict a critical downturn in this specific case.

Lastly, skewness fluctuated sharply, particularly between March and July 2007. After January 2008, these fluctuations diminished and skewness tended to stabilise more, although it remained unstable overall. This could be explained by the fact that, although the crisis culminated in the collapse of Lehman Brothers on 15 September 2008, its initial effects were felt well before that date.

5.4.2 TedSpread

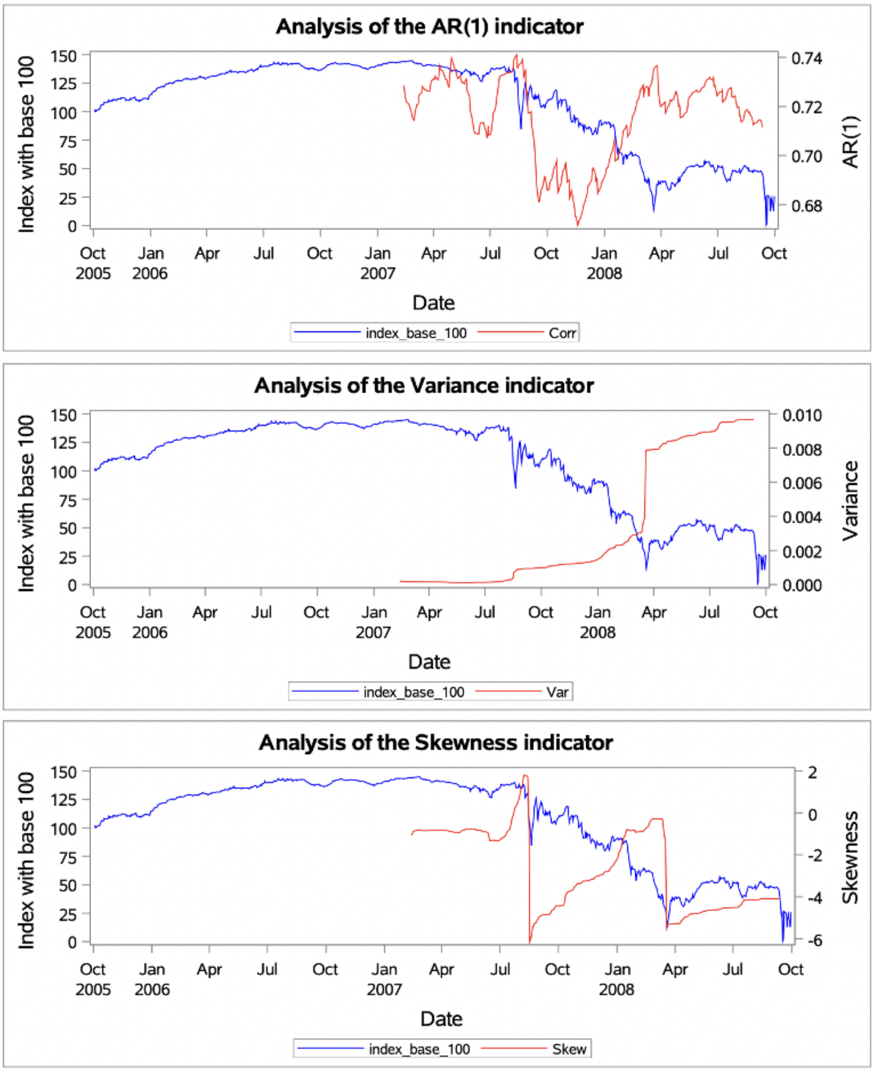

Figure 20: Analysis of indicators for 2008 Financial Crisis with a T/2 window (Ted Spread)

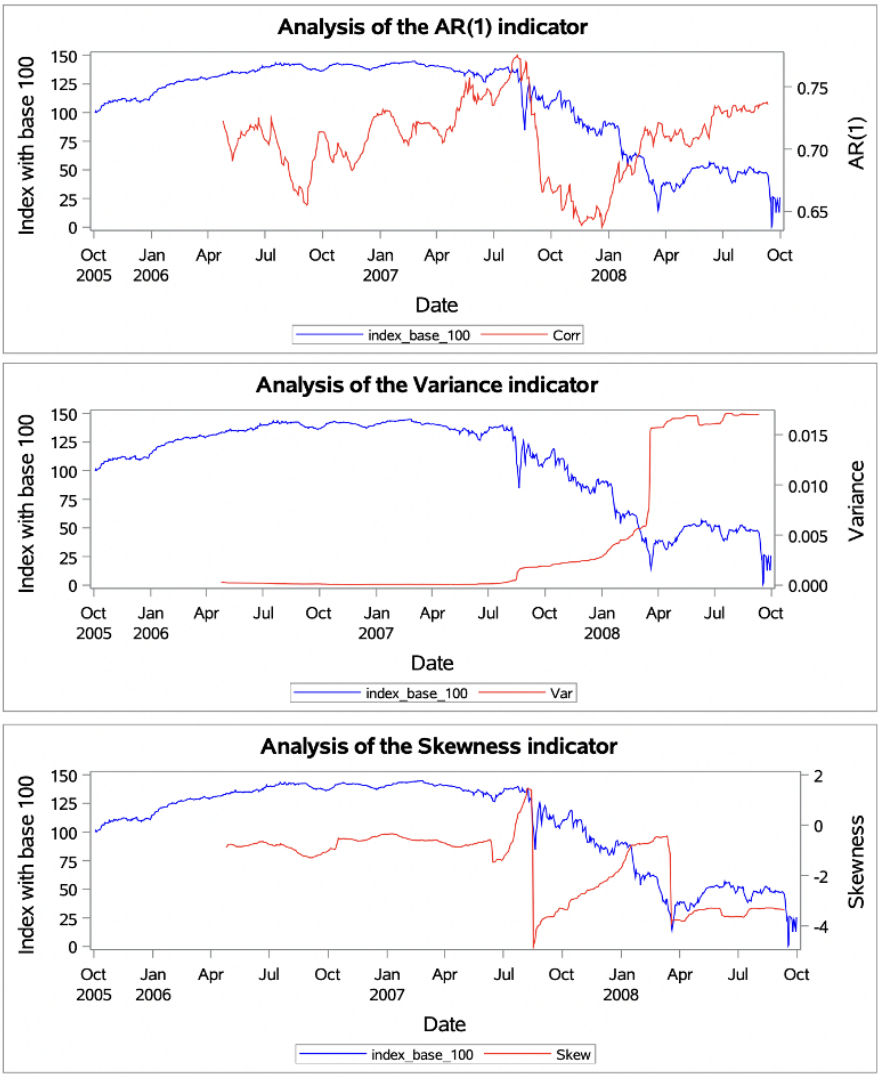

Figure 21: Analysis of indicators for 2008 Financial Crisis with a T/4 window (Ted Spread)

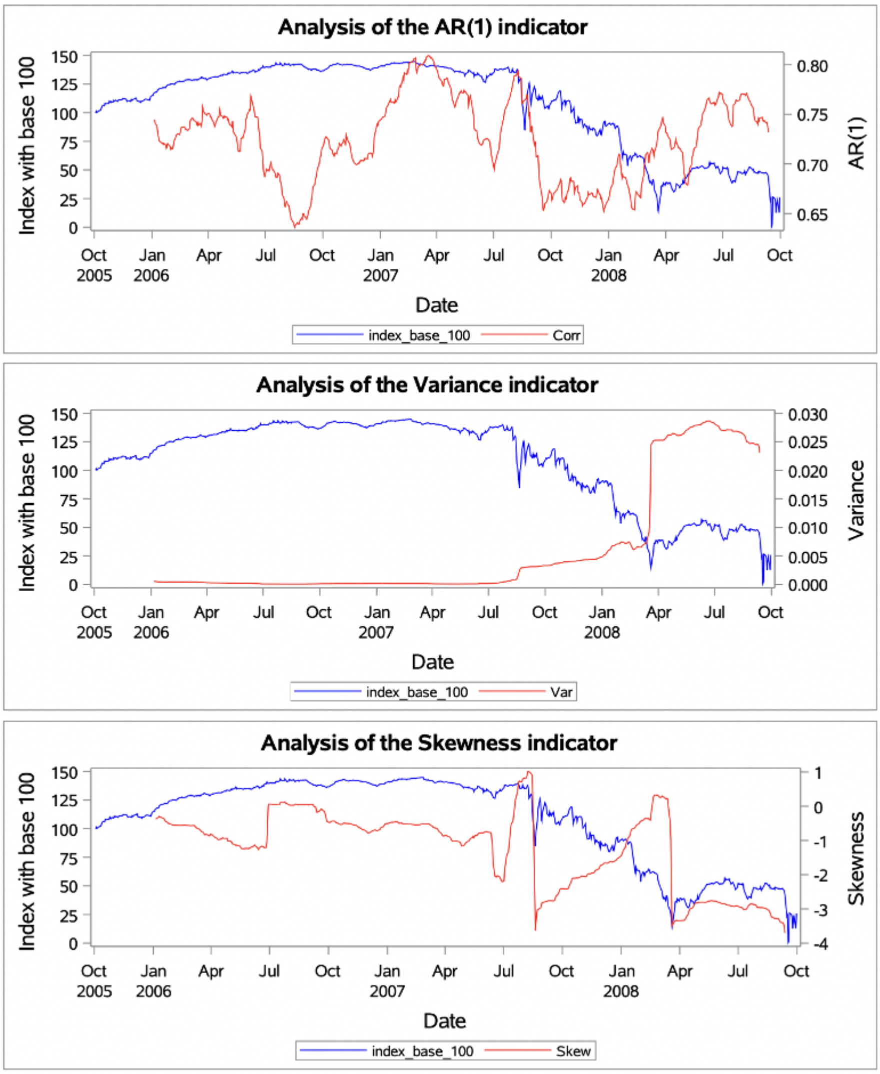

Figure 22: Analysis of indicators for 2008 Financial Crisis with robust parameters (Ted Spread)

When comparing the three different approaches, we notice significant differences in the results. With T/2, the autocorrelation indicator shows only a late rise, after the fall of the index, indicating limited fluctuations due to the smaller window size. This results is an insufficient capture of critical dynamics, making it harder to detect signs of critical slowing down. The variance and skewness show relatively mild trends, and some critical patterns might be missed. In contrast, with T/4, we observe a more pronounced increase in the AR(1) value, indicating greater instability and a stronger link to previous states. The larger window size allows us to capture more fluctuations, revealing clearer signs of approaching instability, although the overall interpretation of the variance and skewness remains similar.

Finally, with the robustness test (we chose the parameters based on Figure 32), the results show even bigger fluctuations and give a clearer view of the system's behavior. The AR(1) indicator changes more sharply, starting at a higher value and showing larger swings, which reflects a more unstable market. The variance reaches a higher peak, showing more instability, and we see a clearer drop back to the initial level. Skewness also shifts more noticeably, confirming the market's instability. This approach, using customized parameters, gives the most reliable and useful results, helping us better understand the system's critical transition.

5.4.3 FTSE100

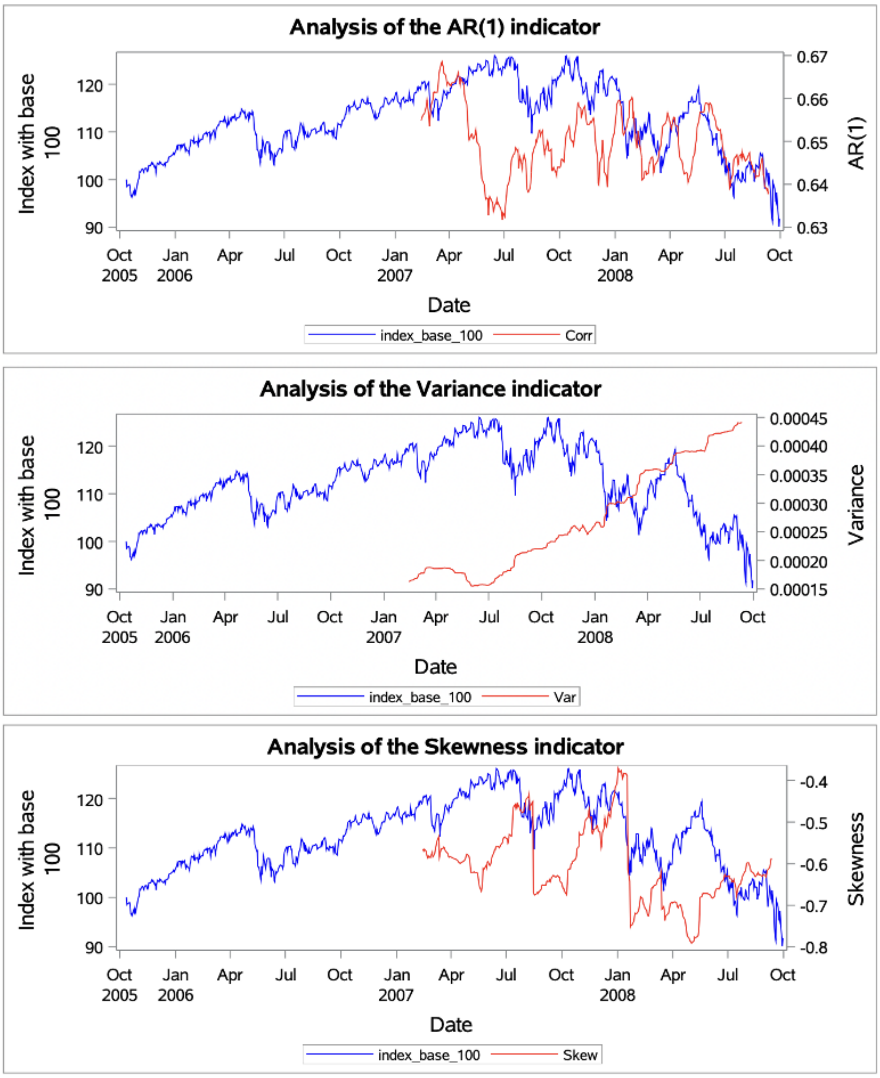

Figure 23: Analysis of indicators for 2008 Financial Crisis with a T/2 window (FTSE 100)

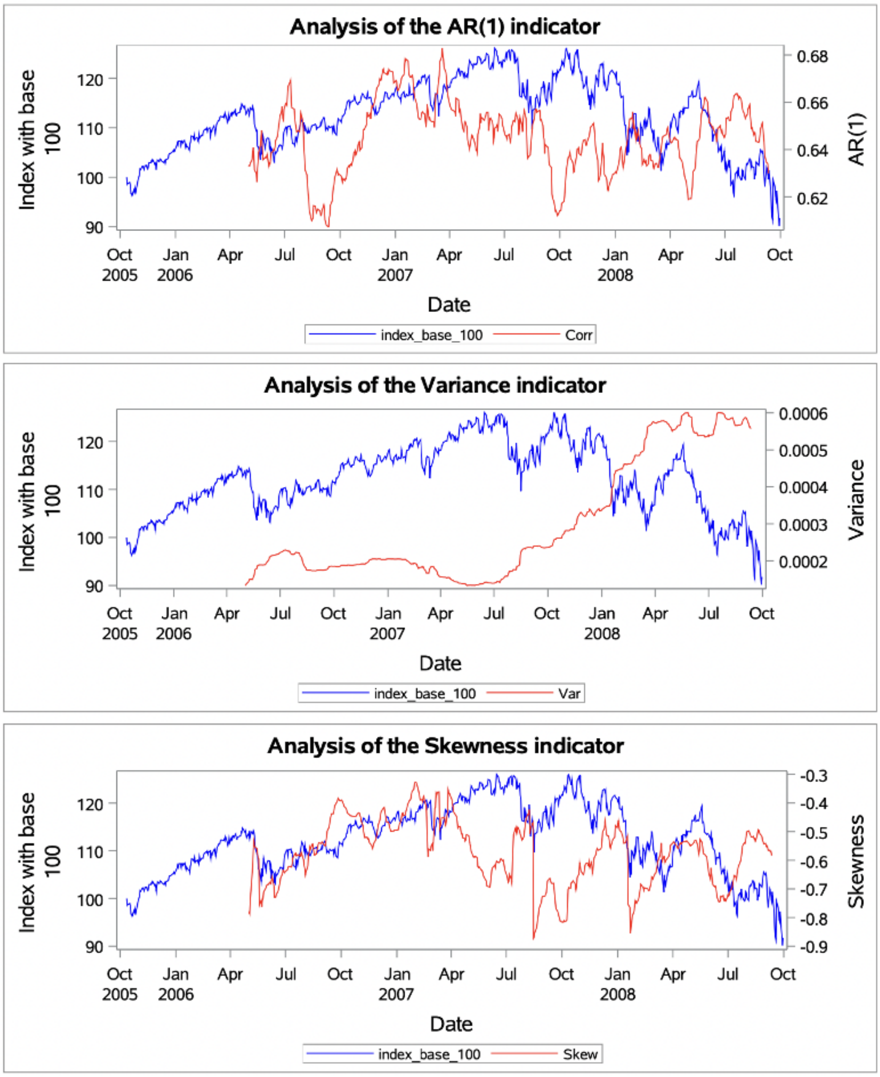

Figure 24: Analysis of indicators for 2008 Financial Crisis with a T/4 window (FTSE 100)

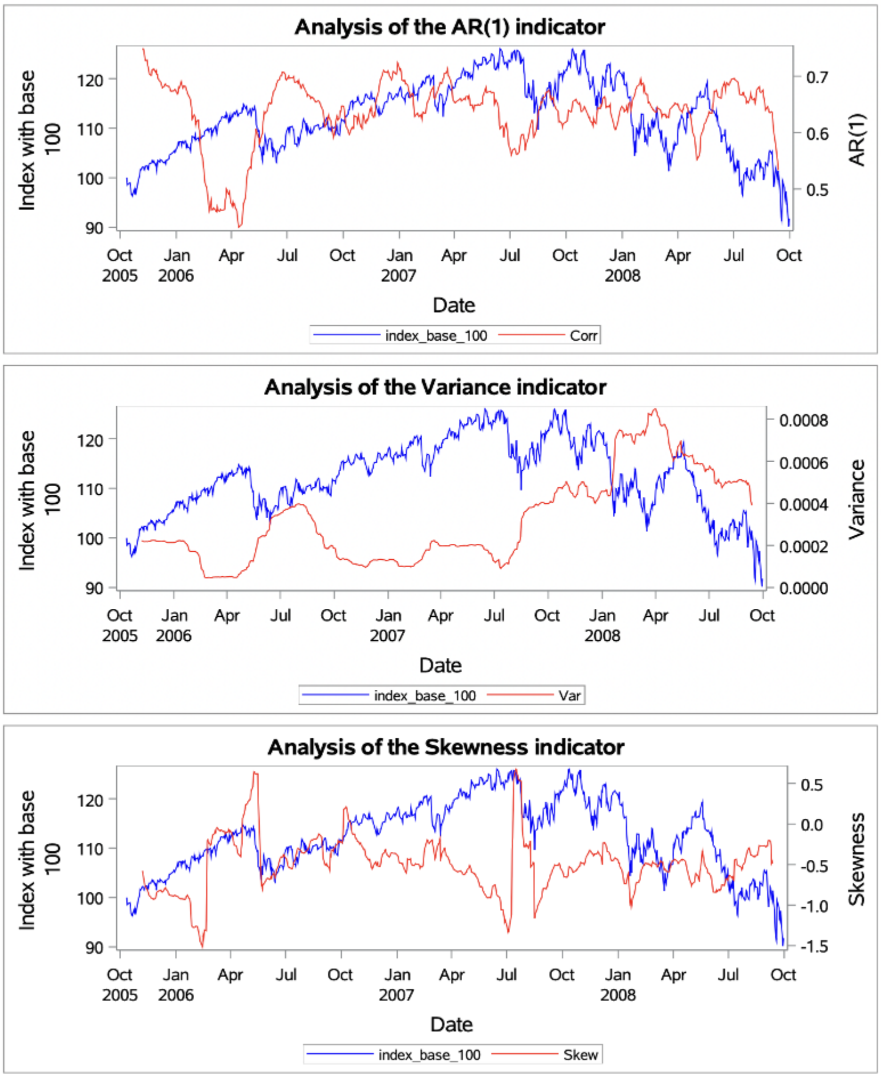

Figure 25: Analysis of indicators for 2008 Financial Crisis with robust parameters (FTSE 100)

Analysis of the AR(1) reveals distinct behaviours depending on the observation method chosen. The T/2 window shows a degree of stability, with values fluctuating in a narrow range between 0.64 and 0.66. This stability contrasts with the greater volatility observed with the robust indicator. One striking point is the significant increase in the AR(1) in May 2008, peaking between June and July, a few months before the collapse of Lehman Brothers. Using the T/2 window, we can see that the AR(1) gradually tends to follow changes in the base 100 index, with this correlation becoming more pronounced as we approach 15 September.The robust indicator, whose parameters are set out in Figure 33), shows a different dynamic : it increases when the index falls. Although it maintained high levels before the crisis, its upward trend began in July 2006, which complicates its interpretation. All in all, this indicator is particularly difficult to interpret, as it does not show a clear trend, whatever the method of analysis used.

As for the variance, whatever the size of the window, it increases relative to its initial state. However, it fluctuates more when the robust indicators are taken into account. According to the CSD theory, the AR(1) and the variance should increase simultaneously. However, validating this theory in this specific case is proving more difficult.

Finally, skewness, which provides additional information on a possible warning of a critical slowdown, remains predominantly negative, with more marked fluctuations when robust indicators are used. There was a significant peak in fluctuations around July 2007, followed by difficulty in stabilising.

5.4.4 Summary of the results of the 2008 crisis

In conclusion, the analysis of the indicators for the 2008 crisis on the various stock market indices shows mixed results that do not allow any definitive conclusions to be drawn immediately. The particular nature of this crisis could explain the lesser effectiveness of CSD indicators compared with previous crises. Indeed, unlike a sudden collapse, the 2008 crisis developed gradually over a period of 12 to 24 months, as shown by the graphs of indices at base 100.

This characteristic of gradual decline, marked by a slow but continuous deterioration in the markets, could explain the limitations of CSD indicators in this context. Traditional warning signals seem less effective at detecting a critical transition when it manifests itself in a succession of small falls rather than a sudden collapse. The gradual deterioration of the financial system, albeit culminating in a major crisis, may have made it more difficult for traditional CSD indicators to detect early warning signs.

This finding raises important questions about the ability of indicators to identify crises that develop over an extended period, perhaps suggesting the need to adapt detection tools to different types of crisis dynamics.

6 Conclusion

The CSD is a promising new approach to crisis prevention, based on a physical model. Its main advantage is its versatility, allowing it to be applied to various types of crisis: climatic, financial or even epileptic, thus going beyond the strictly financial framework that served as the basis for our study.

Our empirical results reveal a certain ambivalence. Early detection of critical transitions in financial markets is more complex than in other dynamic systems. The performance of the indicators varies according to the crises and indices analysed. Variance stands out as the most reliable indicator, while AR(1) presents more nuanced and sometimes even contradictory results, complicating their interpretation. This variability can be explained by the inherent complexity of financial markets, which are subject to multiple external influences and interconnected dynamics.

The 2008 crisis in particular highlighted the limitations of applying the CSD to financial markets. Its gradual development over an extended period, in contrast to more sudden events such as Black Monday, made the detection of precursor signals more delicate. This observation leads us to an important conclusion: CSD is effective in the face of sudden events, but its interpretation becomes more complex in the case of gradual crises. It gives more positive results, especially for Black Monday, but shows mixed results for the Asian Crisis and the Dot-Com Bubble.

For future work, it would be useful to study progressive seizures in greater depth. This could be done by exploring new indicators specific to this type of crisis. For example, it would be interesting to broaden the analysis by exploring higher-order AR(p) autoregressive models, rather than just AR(1). In addition, exploring other indicators not covered in this article could provide additional insights into our understanding of the phenomenon.

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Appendix

A Residuals after smoothing Gaussian Kernel

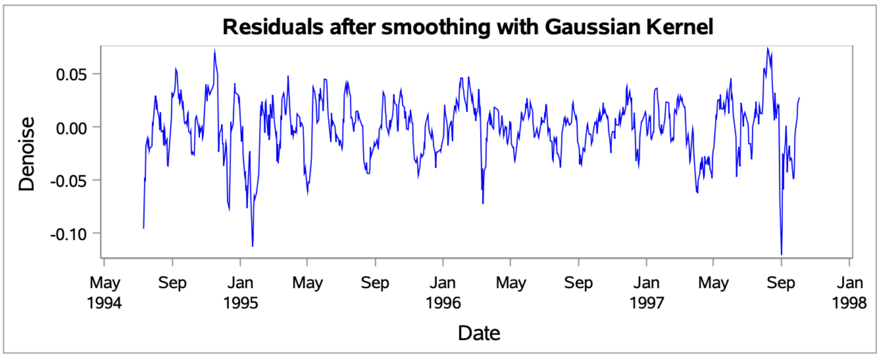

Figure 26: Denoise with $\sigma = 15$ for the Asian Crisis with Hang Seng

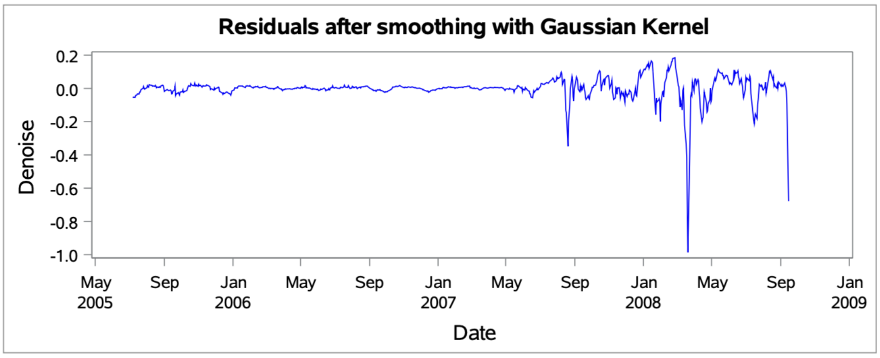

Figure 27: Denoise with $\sigma = 15$ for the 2008 Financial Crisis with TED Spread

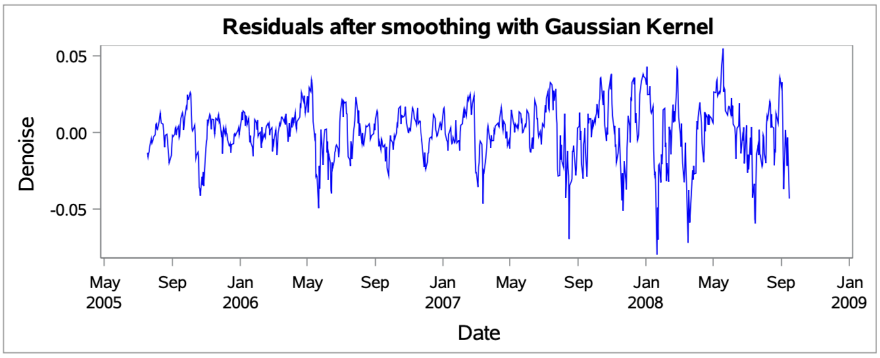

Figure 28: Denoise with $\sigma = 15$ for the 2008 Financial Crisis with FTSE 100

B Contour Plot for the Robustness

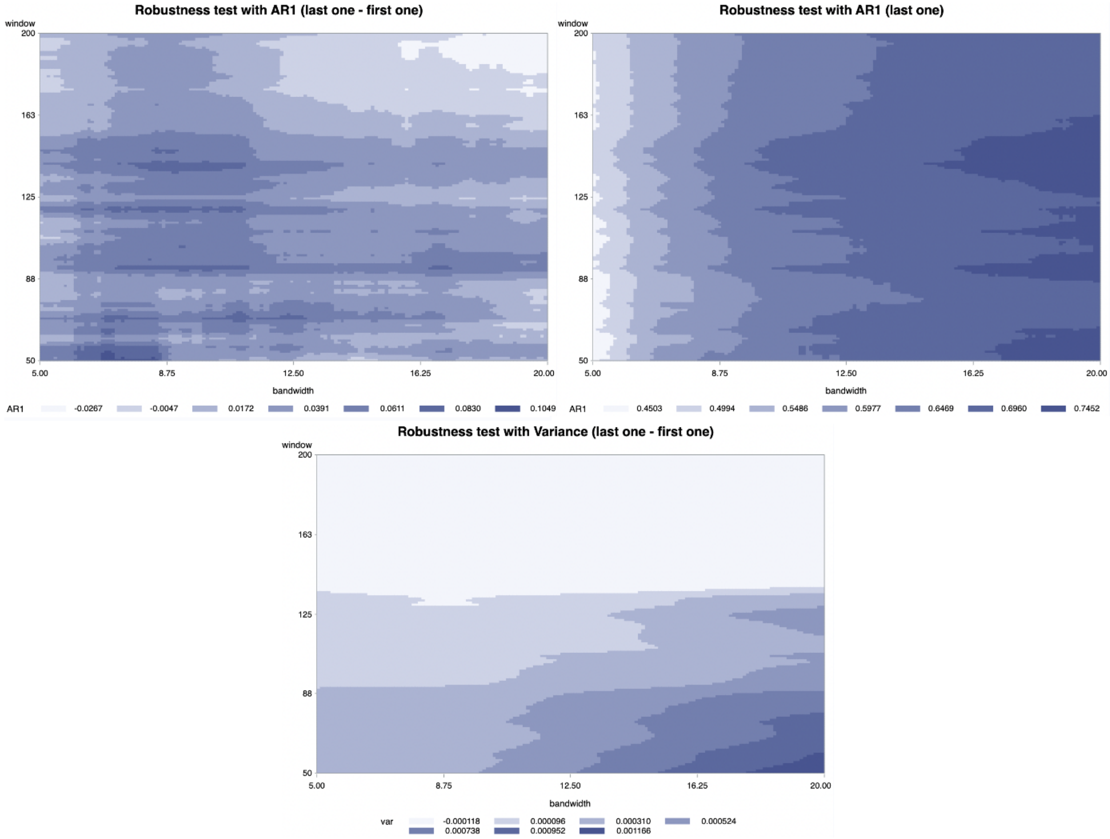

Figure 29: Countour plots for the Asian Crisis

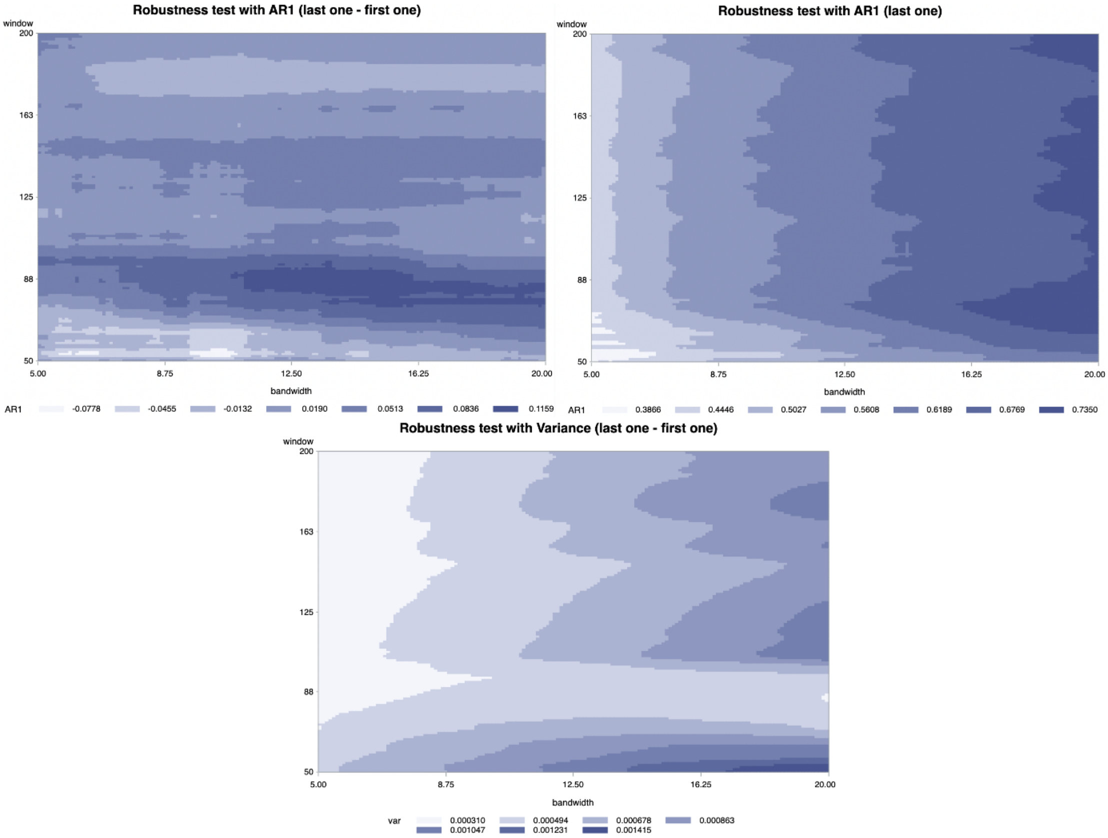

Figure 30: Countour plots for the Dot-com Bubble

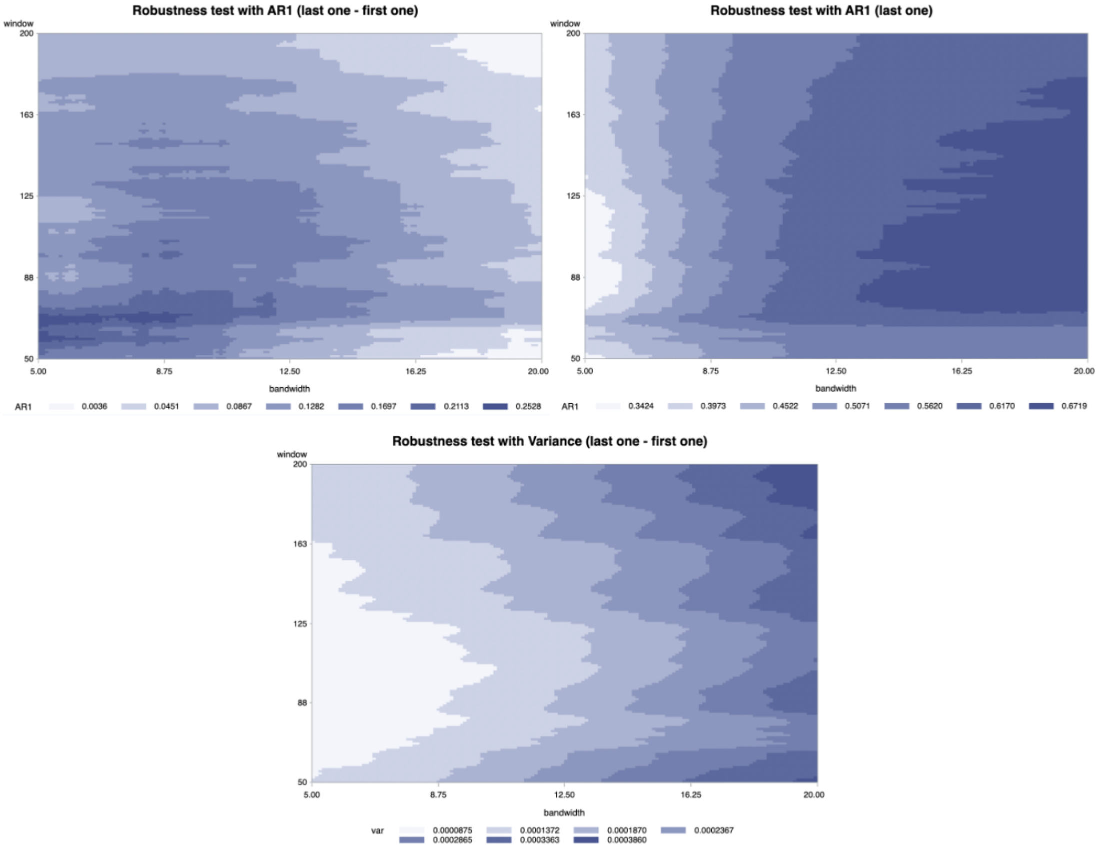

Figure 31: Countour plots for 2008 Financial Crisis with the Ted Spread.

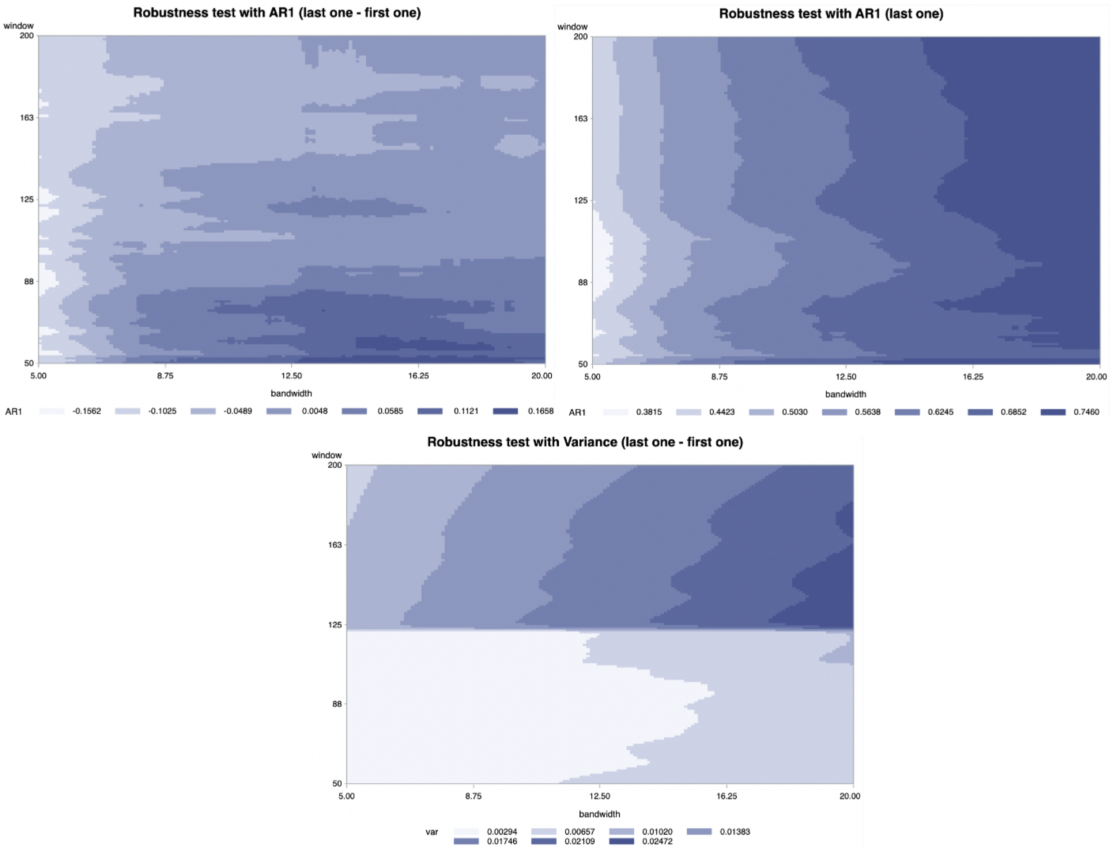

Figure 32: Countour plots for 2008 Financial Crisis with the Ted Spread

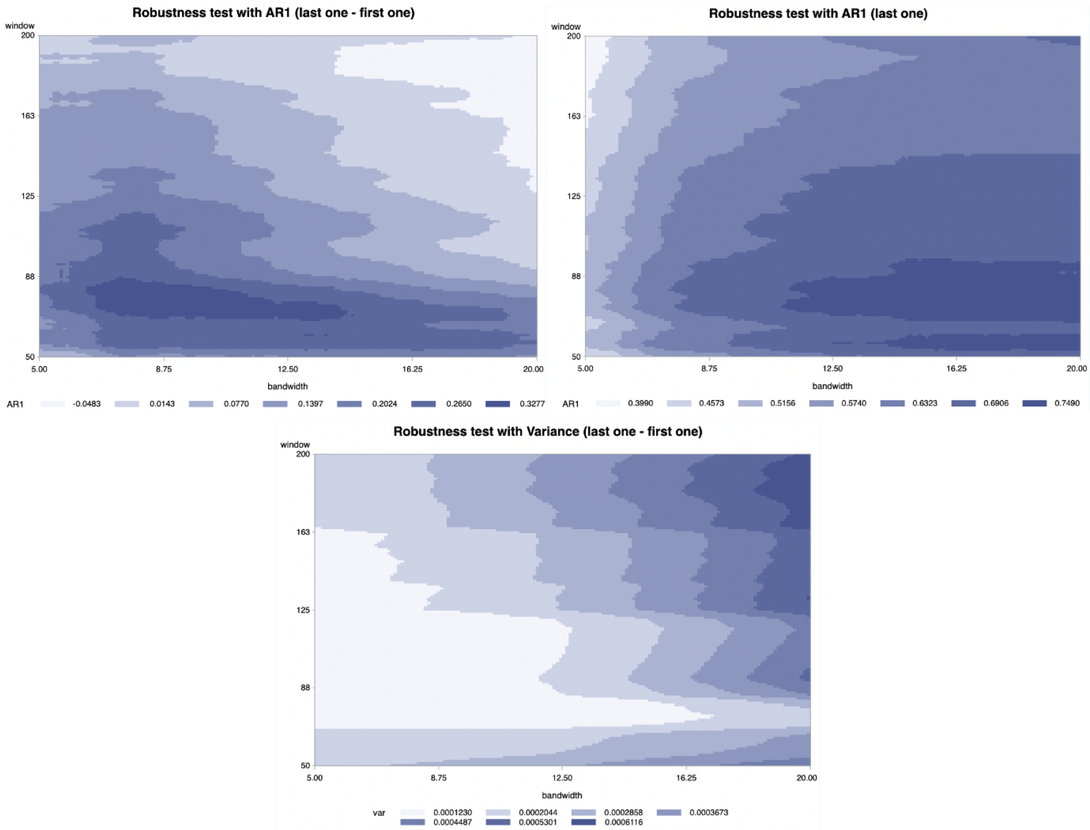

Figure 33: Countour plots for 2008 Financial Crisis with the FTSE 100